AGREEMENT ACROSS COVERS AND COMMON TUNING FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We construct one mass and vacuum pair for the prescribed family of odd reference tori in three-dimensional chamber lattice QED, under explicit primitive and nonlinear-step hypotheses. The pair is uniquely selected among controlled trajectories by zero terminal couplings on the top torus. Signed projection of the primitive kernels and comparison on nonwrapping local lifts identify the actual renormalization steps on smaller odd quotients. A backward mass equation and a forward activity equation give a finite-sequence contraction; continuation and local comparison then transfer the selected pair with uniform bounds.

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1. Introduction

Fix $0 < \rho < 1$, the chamber weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$, and an odd integer $L \geq 4^8$, an integral power M of L with $M \geq L^2$, integers $r \geq 2$ and $p_0, p > 0$, $0 < \varepsilon \leq 1/200$, $1 - \varepsilon < \alpha < 1$, $c_{\mathrm{fl}} \geq 1$, and $0 < \bar{m} \leq 1$. The photon is periodic and the four-component chamber spinors are antiperiodic in all three directions. Each photon site has 48 chamber labels, with measure $\eta_k^3/48$ at mesh $\eta_k = L^{-k}$. For an integer $N \geq 0$, set

$$h = L^{\lfloor \log_L(1/\bar{m}) \rfloor}, \qquad \eta = L^{-N}, \quad K = N + \log_L h, \quad a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j.$$

Here $0 \leq j \leq K$, and $q = e\sqrt{\eta}$ denotes the charge at level zero. We write $\bar{e}_0, \bar{e}_1$ for the fixed charge ceilings, to distinguish them from the running charges e_j. Put $v_j = -\log e_j$, and let r_j be the least power L^i, $i \in \mathbb{Z}_{\geq 0}$, with $r_j \geq v_j^r$. The physical sides are

$$R_{\mathrm{top}} = hMr_K, \quad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^T : T \in \mathbb{N} \cup \{0\},\ hL^T \leq R_{\mathrm{top}}\}.$$

They are independent of N. For $R \in \mathcal{T}(e)$, define $\ell_j = R/a_j$ and $\mathfrak{n}_j = \log_L(\ell_j/(Mr_j))$. The stopping level $k_*(R)$ is the first j with $\mathfrak{n}_j = 0$, when it exists. Write $k_{\mathrm{top}} = k_*(R_{\mathrm{top}})$. A star denotes evaluation at $k_*(R)$: for example $a_* = a_{k_*(R)}$ and $\mu_* = \mu_{k_*(R)}$.

Numerical hypotheses. Assume

$$0 < e\sqrt{h} \leq \bar{e}_1 \leq \min(\bar{e}_0, 1/3), \qquad \bar{e}_0^{\varepsilon} \leq 1/4,$$
$$c_{\mathrm{fl}} L^{-3/8+6/200} \leq 1/16, \qquad L^{-5/8} \leq 1/4,$$

and, for every $v \geq -\log \bar{e}_1$,

$$v \geq 8, \qquad \left(1+\frac{3\log L}{2v}\right)^r < L, \qquad v^r > 31L+16,$$
$$(c_{\mathfrak{fl}}+2)e^{-\varepsilon v} \leq \frac{1-L^{-\varepsilon/2}}{16}. \tag{1}$$

The activity maps. Use the real chamber activity spaces, with independent connection arguments, defined in section 2. Write $\mathscr{E}_j$ for an activity with external connection $\mathscr{C}$, and $E_j = \mathscr{E}_j|_{\mathscr{C}=0}$. At every used scale impose all the simultaneous hypotheses (i)–(iv) of [NL, Theorem A], including its primitive, inverse-domain, determinant, extraction, reconstruction, contour and connected-logarithm conditions. The constants, norms and operators are those of that theorem and of [NC, Theorems A–B]; their definitions and the parts used below are recalled in sections 2 and 3. These are hypotheses on the actual chamber maps. In particular the following tuning inequalities do not replace their domain or numerical conditions. The decay weight satisfies $\kappa \geq 7\kappa_0 + 7$, where $\kappa_0 = 54\log 72$; the constants are calibrated together as in the cited step theorem.

On the top torus choose the photon and fermion families of section 4.3, with polymer decay $\kappa_w + \kappa_0$, where $\kappa_w = \kappa + \kappa_0 + 3$. On each smaller odd quotient use their projections, with common constant $C_w = 64K_{\mathrm{pol}}C_w^{(1)}$, where $K_{\mathrm{pol}} = \exp(\kappa_0)/36$ is the scalar polymer-sum constant of [NC, Appendix B, (81)]. Both incoming and outgoing sides are required to have $\ell/(LM)$ odd and at least 13. The step comparison requires $\ell_{\mathrm{small}}/L > (31L+16)M$; the stopping calculation verifies these side conditions at all steps used in the theorem.

For a comparison with smaller unit side ℓ_1, a connected covering polymer $\widetilde{X}$ is admissible when its smallest rectangular box has largest side at most $\ell_1 - 14(2L+1)M$. Its two torus images are called flat. Local agreement means equality under the common covering generators for connection pairs agreeing on $\mathrm{box}(\widetilde{X})^{\sim 6(2L+1)}$. Here the superscript denotes enlargement by that number of M-cube layers. Proposition 5.1 justifies this identification.

The estimates used for tuning take the following form. Set $w_j^{\mathrm{m}} = e_j^{3/4-8\varepsilon}$ and $w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}$. The no-slot data norm is $\max\{|m|/w_j^{\mathrm{m}}, \|E\|_j/w_j^{\mathrm{E}}\}$; with an external slot, replace $\|E\|_j$ by $\|\mathscr{E}\|_j^x$. Radius-four data balls mean the corresponding closed balls in the complexified spaces. Unit balls below are closed balls in their real subspaces. At zero the activity coordinate is fixed to zero. The maps preserve reality and obey

$$m_{j+1} = L(m_j + m(E_j)), \qquad \varepsilon_{j+1}^v = L^3(\varepsilon_j^v + \varepsilon_v(E_j) + \delta\varepsilon_j^0), \tag{2}$$
$$E_{j+1} = \Phi_j(m_j, E_j), \qquad \Phi_j(m,E) = \mathcal{L}(RE)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m,E). \tag{3}$$

Restriction to $\mathscr{C} = 0$ commutes with the full external-slot recursion and does not increase its norm. Scalar extraction is linear and real,

$$|m(E)| \leq c_{\mathfrak{fl}} e_j^{1/2}\|E\|_j, \qquad |\varepsilon_v(E)| \leq c_{\mathfrak{fl}}\|E\|_j, \qquad 0 \leq \delta\varepsilon_j^0 \leq e_j^7.$$

On the unit ball the normalized size and difference estimates are

$$\begin{aligned} L^{-1}w_{j+1}^m/w_j^{\mathrm{m}} &\leq 1/4, \\ \|\Phi_j(m,E)\|_{j+1}/w_{j+1}^E &\leq \tfrac{1}{16}\|E\|_j/w_j^{\mathrm{E}} + 2e_{j+1}^{\varepsilon}, \\ |m(E)-m(\widetilde{E})|/w_j^{\mathrm{m}} &\leq c_{\mathfrak{fl}} e_j^{\varepsilon} d_E, \\ \|\Phi_j(m,E)-\Phi_j(\widetilde{m},\widetilde{E})\|_{j+1}/w_{j+1}^E &\leq \tfrac{1}{16}d_E + e_{j+1}^{\varepsilon}(d_m + d_E), \end{aligned} \tag{4}$$

where $d_m = |m-\widetilde{m}|/w_j^{\mathrm{m}}$ and $d_E = \|E-\widetilde{E}\|_j/w_j^{\mathrm{E}}$. The same size estimate holds for the external-slot norm. The fixed-point argument uses the no-slot difference estimate. Initially $E_0 = \mathscr{E}_0 = \mathfrak{P}_0 = 0$, where $\mathfrak{P}_j$ is the scalar external family in section 3. The first step uses the chamber definitions in section 3.

Theorem A (A common mass and vacuum pair)**.** Under the hypotheses above, the following assertions hold. For each $R \in \mathcal{T}(e)$ there is a finite threshold $N_{\min}(R,e)$ such that the first stop exists for $N \geq N_{\min}(R,e)$. Its physical quantities are then independent of N. The thresholds are nonincreasing and the stopping levels nondecreasing with R; their precise formulas are in (52).

For $N \geq N_{\min}(R_{\mathrm{top}},e)$, there is exactly one real pair $P_N(e) = (m_{0,N}(e), \varepsilon^v_{0,N}(e))$ whose top-torus trajectory is controlled and has $m_{k_{\mathrm{top}}} = \varepsilon^v_{k_{\mathrm{top}}} = 0$. Controlled means that each preceding step is defined in its radius-four data ball and $|m_j| \leq w_j^{\mathrm{m}}$ through the stop. For every $R \in \mathcal{T}(e)$ with $N \geq N_{\min}(R,e)$, the same pair is controlled on R. Up to its stop, the scalar mass and vacuum sequences equal the top sequences, while the activities and external scalar families agree on admissible flat lifts under the algebra identification of proposition 5.1. On each such torus,

$$|m_j| \leq \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \leq \|\mathscr{E}_j\|_j^x \leq 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^v| \leq 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}. \tag{5}$$

The smaller terminal couplings need not vanish.

Within the fixed construction, the pair is determined by $(N,e,\bar{m})$ and is independent of the smaller side, twist, external connection and probe. Define only $m_{0,N}(e) = 0$ below the top definition threshold. For every $N \geq 0$,

$$m_{0,N}(e) \in \mathbb{R}, \qquad |m_{0,N}(e)| \leq \tfrac{1}{8} q^{3/4-8\varepsilon} < \tfrac{1}{8}.$$

The total bare mass $\boldsymbol{\mu}_N = \bar{m}L^{-N} + m_{0,N}(e)$ has no asserted sign. On R^*, for $N \geq N_{\mathrm{ref}}(e) := N_{\min}(R^*,e)$,

$$a_* \in \{h/L, h/L^2\}, \qquad \ell_* = Mr_*, \qquad r_* > 31L + 16, \qquad L^{-3} < \mu_* \leq L^{-1}.$$

■□

The theorem concerns the prescribed family of odd reference tori. The scalar pair is selected before inserting a smaller side, a harmonic twist or an external probe. The proof has two parts. On the top torus, the mass equation is solved backwards from zero terminal mass, while the activity equation runs forwards from zero initial activity. In the weighted sequence norm these two equations define a contraction with factor 5/16. The vacuum is recovered by an explicit backward sum.

To use that pair on a smaller torus, we compare the actual local operations. Odd covers preserve the antiperiodic character. Projection gives a signed finite image sum; when weakening parameters confine a row to a common nonwrapping lift, only one image survives. The local Grassmann algebra then identifies extraction, determinant traces, both finite localizations and the fluctuation integrals. Absolute convergence passes this equality to the connected logarithm. At each smaller-volume step, its own continuation estimate controls the full activity norm, and local comparison identifies the scalar couplings with those of the top trajectory.

The cover and tuning arguments are developed in [QED, §§15–16]. The photon, fermion, extraction and nonlinear constructions are supplied by [MB; FB; NC; NL]. We give the cover comparison and finite-sequence tuning proofs here, with those constructions as explicit inputs.

The definitions in section 2 and the input identities in section 3 fix the maps compared in sections 4 and 5. The stopping calculation is in section 6; section 7 proves the common-pair theorem.

2. Activities, connections and coefficient norms

We specify the local activity space used in theorem A. These are the conventions of [NC, §2].

2.1. Chamber sites, pairs and connection domains.

Write O_{cub} for the signed permutation matrices on $\mathbb{R}^3$. A chamber site is $x = (c,g)$, with photon projection $\pi x = c$ and $g \in O_{\mathrm{cub}}$. Integration over chamber sites uses $\mathrm{d}x = (\eta_k^3/48)\sum_{c,g}$. We take M-cubes centered at $M\mathbb{Z}^3/\ell\mathbb{Z}^3$; their lattice sites form a partition. Blocks of every odd side used below are centered in the same way.

For the three internal neighbors let s_0 change the sign of the third coordinate, let s_1 exchange the second and third coordinates, and let s_2 exchange the first and second. Let r_{e_1} change the sign of the first coordinate. Set

$$x^{(i)} = (c, gs_i) \quad (0 \le i \le 2), \qquad x^{(3)} = (c + \eta_k g e_1, g r_{e_1}).$$

Each neighbor map is an involution. An oriented bond field B is antisymmetric under reversal and periodic on the photon lattice. Write $B_\mu(c) = B(c, c + \eta_k e_\mu)$, and use the forward differences

$$\partial_\nu B_\mu(c) = \eta_k^{-1}(B_\mu(c + \eta_k e_\nu) - B_\mu(c)), \qquad (d\lambda)_\mu(c) = \eta_k^{-1}(\lambda(c + \eta_k e_\mu) - \lambda(c)).$$

Thus $(dB)_{\nu\mu} = \partial_\nu B_\mu - \partial_\mu B_\nu$. Distances between photon sites are sup distances, using the shortest lift when a distance bound is imposed. For $0 < \alpha < 1$ the three norms are

$$\|B\|_1 = \|B\|_\infty, \qquad \|B\|_2 = \|\partial B\|_\infty, \qquad \|B\|_3 = \|\delta_\alpha \partial B\|_\infty = \sup_{\substack{\mu,\nu,c,c' \\ 0<d(c,c')\le 1}} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c,c')^\alpha}.$$

Local versions use only the indicated cells in the given region. Let $\square^\dagger$ be the concentric enlargement of an M-cube to side $(2L+1)M$. Put $\theta_i = e_k^{-3/4+i\varepsilon}$. The connection domains are

$$\begin{aligned} R_k &= \{B : \|B\|_i < \theta_i,\ 1 \le i \le 3\}, \\ \widetilde{R}_k &= \{B_0 + B_1 : B_1 \in R_k,\ B_0 \text{ real, and on each } \square^\dagger, \\ &\qquad\qquad B_0 = B^{\mathrm{b}}_\square + d\lambda_\square \text{ with } B^{\mathrm{b}}_\square, \lambda_\square \text{ real and } \|B^{\mathrm{b}}_\square\|_i < \theta_i\}, \\ \mathfrak{X}_k &= \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_k^{-1+2\varepsilon},\ \|\partial\mathscr{C}\|_\infty < e_k^{-1+3\varepsilon}\}. \end{aligned} \tag{6}$$

Only existence of these representatives is required. For $0 < c \le 1$, the notation $c\widetilde{R}_k$ multiplies all three bounds for both summands by c. For $\widetilde{R}_k^-$ the local real bounds remain θ_i, while the global summand has bounds $(1 - e_k^\varepsilon)\theta_i$.

For a short ordered pair $x = (c, g), x' = (c', g')$, let $\tau B(c, c')$ be the arithmetic mean of the sums of B along the six coordinate-order paths from c to c'. The bond values in this sum are unscaled; define

$$U_B(x, x') = \exp\{i e_k \eta_k \tau B(c, c')\}, \qquad (\delta_{\alpha,B}\psi)(x, x') = \frac{U_B(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta_k)^\alpha}.$$

For pairs within one cluster the transport is one. Barred fields use U_B^{-1}. These formulas specify evaluation; the pair generators in the algebra remain independent.

We use Pauli matrices and

$$\gamma_\mu = \sigma_3 \otimes \sigma_\mu, \quad \gamma_5 = \sigma_1 \otimes 1_2, \quad \Sigma_3 = \sigma_3 \otimes 1_2, \qquad \mathsf{C} = i\gamma_2, \quad \varkappa = \gamma_5 \mathsf{C} = \sigma_2 \otimes \sigma_2.$$

Here C is real with $\mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}$, while $\varkappa$ is a real symmetric involution. The spin group below is the group generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_5$. Its diagonal coordinate action $r(S)$ is determined by $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Connections transform by pullback on oriented bonds.

2.2. **The local Grassmann algebra.** Fix a level k, mesh $\eta_k = L^{-k}$ and an admissible side ℓ, with $\ell/(LM)$ odd and at least 13. Let $\mathcal{P}_k$ be the collection of face-connected polymers of M-cubes on this torus. Write $|X|_M$ for the number of its M-cubes. The number $Md_M(X)$ is the infimum of the total Euclidean length of a connected finite net of straight segments meeting every closed cube of X; the infimum includes the segment representations on the torus. The notation $X^{\sim n}$ means enlargement by n layers of M-cubes in the sup metric. A polymer is small when $d_M(X) \le L$. A small polymer has an injective lift, unique up to a period translation, by the diameter and lifting estimates of [NC, Appendix B]. Write $\mathrm{Vol}(X) = M^3|X|_M$.

The independent Grassmann generators are ψ, $\bar\psi$ on the chamber sites $x = (c, g)$ and χ, $\bar\chi$ on ordered pairs. For distinct chamber sites on the covering lattice, put $d_X(x, x') = \max(|c - c'|_\infty, \eta_k)$ and restrict pairs to $d_X \le M$. Pairs on the torus are simultaneous period classes; their shortest lifts

are unique since $2M < \ell$. A pair is admissible for X when its photon sites and all six rectilinear paths between them lie in X, or when it is $(x, x^{(3)})$ and one of its photon sites belongs to X. Thus a pair may leave X only along a bond of type 3. Pairs inside a cluster have $d_X = \eta_k$ and are admissible when their common photon site belongs to X. Ordinary generators acquire the sign $(-1)^{n_1+n_2+n_3}$ under translation by ℓn. Pair generators acquire the same sign when both endpoints are translated by ℓn. The kernels are antiperiodic in each ordinary argument and in each simultaneous pair argument with these conventions.

Let $\xi = (x, \beta, \omega)$ index an ordinary leg and $\zeta = (x, x', \beta, \omega)$ a Hölder leg, where $\omega = 0$ is unbarred and $\omega = 1$ is barred. We use the index measures

$$\nu(\xi) = \eta_k^3/48, \qquad \nu(\zeta) = (\eta_k^3/48)^2.$$

Integrals sum the spin and bar indices as well. An activity has a unique expansion with separately antisymmetric kernels,

$$\mathscr{E}(X; \mathscr{A}, \mathscr{C}) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi \, \mathrm{d}\zeta \, \mathscr{E}_{nm}(X; \mathscr{A}, \mathscr{C}; \xi; \zeta) \psi(\xi_1) \cdots \psi(\xi_n) \chi(\zeta_1) \cdots \chi(\zeta_m). \tag{7}$$

Barred generators are included in this notation. The generator weights, as distinct from the index measures, are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}.$$

For an element F the weighted coefficient norm is

$$\|F\|_{h_k} = \sum_{n,m} \frac{h_{k,1}^n h_{k,2}^m}{n!m!} \int \mathrm{d}\xi \, \mathrm{d}\zeta \, |F_{nm}|. \tag{8}$$

In particular a generator has norm $h_{k,i}$; the weight ν enters only when coefficients are written as integral kernels.

An activity family in $K_k^{x,\mathbb{C}}$ is balanced: every monomial has as many barred as unbarred legs. Its kernels are holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, vanish unless all ordinary legs lie over X and all pairs are admissible for X, and read the two connections only on bonds meeting X. The norm is

$$\|\mathscr{E}\|_k^x = \sup_{X \in \mathcal{P}_k} \exp(\kappa d_M(X)) \sup_{(\mathscr{A}, \mathscr{C}) \in \widetilde{R}_k \times \mathfrak{X}_k} \|\mathscr{E}(X; \mathscr{A}, \mathscr{C})\|_{h_k}. \tag{9}$$

We require $\kappa \geq \kappa_0$ and finite norm.

The required symmetries act on both the connections and the generators. For $\tilde{g} = (S, a)$ with the coordinate sign action $r(S)$ and $a \in M\mathbb{Z}^3$, put

$$\tilde{g}(c, g) = (r(S)c + a, r(S)g), \qquad a_{\tilde{g}}\psi(x) = S\psi(\tilde{g}^{-1}x), \qquad a_{\tilde{g}}\bar{\psi}(x) = (S^{-1})^T \bar{\psi}(\tilde{g}^{-1}x).$$

The same spin matrices act on pair generators after transforming both endpoints. Local gauge transformations multiply an unbarred ordinary or pair leg by $\exp(ie_k\Lambda(c))$, using the photon site of its first point, and a barred leg by the inverse phase. Write this automorphism as G_Λ. Charge conjugation and the antilinear reality map act by

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \qquad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \qquad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

and by the same rules on the Hölder generators.

The covariance requirements on a family are

$$\begin{aligned} a_{\tilde{g}}\mathscr{E}(\tilde{g}X; \tilde{g}\mathscr{A}, \tilde{g}\mathscr{C}) &= \mathscr{E}(X; \mathscr{A}, \mathscr{C}), \\ \mathrm{Cg}\,\mathscr{E}(X; -\mathscr{A}, -\mathscr{C}) &= \mathscr{E}(X; \mathscr{A}, \mathscr{C}). \end{aligned} \tag{10}$$

For real scalars λ, λ' and two pairs in the domain which satisfy $\mathscr{A}^1 = \mathscr{A} - d\lambda$ and $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on bonds at distance at most one from X, the local gauge requirement is

$$G_{\lambda+\lambda'}\mathscr{E}(X; \mathscr{A}^1, \mathscr{C}^1) = \mathscr{E}(X; \mathscr{A}, \mathscr{C}).$$

The real subspace K_k^x has in addition

$$\mathbf{K}\mathscr{E}(X;-\overline{\mathscr{A}},-\overline{\mathscr{C}})=\mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

All requirements refer to the generators before evaluating fields. They are essential to the form of the counterterms.

The spaces $K_k^{\mathbb{C}}, K_k$ without the second connection argument have the same definitions on $\widetilde{R}_k$. Restriction $\mathscr{E}|_0(X;\mathscr{A})=\mathscr{E}(X;\mathscr{A},0)$ maps $K_k^{x,\mathbb{C}}$ into $K_k^{\mathbb{C}}$ and has norm at most one. The real subspaces also restrict. Analyticity, locality and the symmetries follow by setting the second argument and its gauge function to zero; the norm bound follows from the smaller supremum.

2.3. **Total connections.** The comparison domain $\widetilde{G}_k=\widetilde{\mathcal{G}}_k$ consists of fields B_0+B_1 with B_0 real and $\|B_1\|_\infty<e_k^{-1+\varepsilon}$, where on each $\square^{\sim 5}$ the field B_0 has a real-gauge representative of supremum norm below $e_k^{-1+\varepsilon}$. Here $\square^{\sim 5}$ has the same center and side $11M$. The real-gauge representatives in the domains defined above may be chosen independently on each enlarged cube.

Assume $L\ge 5$, $0<\varepsilon\le 1/200$ and

$$e_k^{\varepsilon}\le \tfrac14,\qquad e_k^{1/4}+e_k^{\varepsilon}\le 1,\qquad 4e_k^{1/4+2\varepsilon}+2e_k^{3\varepsilon}\le 1. \tag{11}$$

Then

$$\widetilde{R}_k+\mathfrak{X}_k\subset\widetilde{G}_k. \tag{12}$$

Indeed, write $\mathscr{A}=A_0+A_1$ with $A_1\in R_k$ and A_0 locally real-gauge equivalent on $\square^\dagger$ to $A^b_\square$ with the three threshold bounds. Restrict the representative to $\square^{\sim 5}\subset\square^\dagger$. Its supremum norm is below $e_k^{-3/4+\varepsilon}\le e_k^{-1+\varepsilon}$, and

$$\|A_1+\mathscr{C}\|_\infty<e_k^{-3/4+\varepsilon}+e_k^{-1+2\varepsilon}=(e_k^{1/4}+e_k^{\varepsilon})e_k^{-1+\varepsilon}\le e_k^{-1+\varepsilon}.$$

This is the required decomposition of the total connection. Real gauge transformations can absorb any real constant part, locally on the nonwrapping cubes.

Put $\mathfrak{i}(B)=e_k\|\operatorname{Im}B\|_\infty$ and $q_d(B)=e_k\|dB\|_\infty$. The same decomposition gives

$$\mathfrak{i}(\mathscr{A}+\mathscr{C})\le e_k^{1/4+\varepsilon}+e_k^{2\varepsilon}\le 1,\qquad q_d(\mathscr{A}+\mathscr{C})\le 4e_k^{1/4+2\varepsilon}+2e_k^{3\varepsilon}\le 1. \tag{13}$$

For the curvature, every plaquette lies in an enlarged cube, and $dA_0=dA^b_\square$ there. This costs at most $2\|\partial A^b_\square\|_\infty$; the global small field costs another $2\|\partial A_1\|_\infty$, and the external field costs $2\|\partial\mathscr{C}\|_\infty$. The imaginary bound uses only that A_0 is real.

The domains are open: the strict inequalities define open sets for the global summand, which can absorb a sufficiently small complex perturbation while the local real representative is kept fixed. The norm-defined sets are invariant under sign and conjugation; the local real-gauge definitions are also invariant under real gauges and real constant additions. Coordinate sign changes and M translations preserve them. Their derivative and Hölder norms are preserved, with a possible reversal of a bond and translation of its base point. Thus (12) applies also at the signs, conjugates and real gauges used below.

2.4. **The operator and photon data.** The injection and cluster mean are $(Jv)(c,g)=v(c)$ and $(J^*f)(c)=48^{-1}\sum_g f(c,g)$. The normalized average $\mathcal{Q}_k=Q_kJ^*$, its reverse $\mathcal{Q}_k^\sharp=\mathcal{Q}_k(-B)^T$, the one-step average Q, and the chamber operator D_B are given in [FB, §1]. Put $b_k=(1-L^{-1})/(1-L^{-k})$ for $k\ge 1$. At these levels set

$$S_k(B)=(D_B+\mu+b_k\mathcal{Q}_k^\sharp\mathcal{Q}_k)^{-1},\quad H_k(B)=b_kS_k(B)\mathcal{Q}_k^\sharp(B),\quad D_k(B)=b_k-b_k^2\mathcal{Q}_k(B)S_k(B)\mathcal{Q}_k^\sharp(B).$$

The one-step projection is $P_L(B)=Q^T(-B)Q(B)$. The barred map is $\bar H_k(B)=\mathsf{C}^{-1}H_k(-B)\mathsf{C}$. At $k=0$, $H_0=\bar H_0=1$ and $D_0(B)=D_B+\mu$. The symbol $\mathcal{S}(a,\ell)$ denotes the four-component spinor space on the a-spaced torus, with measure a^3; the superscript ch adds chamber labels and divides

the measure by 48. All spaces have the antiperiodic convention stated above. For later use, the hopping notation of [FB, §2.1] is

$$R_i(B)f(x) = U_B(x, x^{(i)})f(x^{(i)}), \qquad \mathcal{L}_B = \sum_{i=0}^{3} w_i(1 - R_i(B)), \qquad M_i(c, g) = \tfrac{w_i}{2}(\rho + \gamma \cdot g\mathbf{n}_i),$$

where $(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6)$ and $(\mathbf{n}_0, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1)$. Thus $\widehat{d}_i(c, g) = -g\mathbf{n}_i$. On a nearest-neighbor photon bond write $u_B(c, c') = \exp(ie_k\eta_k B(c, c'))$; this is the transport $U_B(x, x^{(3)})$ when c, c' are its endpoints.

For photon fields use the Landau reconstruction H_k^{ph}, the constrained minimum $A_k^{\min}$ and the real coordinate map $C : V_k \to \mathcal{Z}$ of [MB, Theorem A]. Write $C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}$ and $\mathrm{Sq}_k = CC_k^{1/2}$. The coordinate basis $\{\Upsilon\}$ is orthonormal and has the specified block or face-bond supports. The law μ_I is the product standard Gaussian in these coordinates. For the fixed exponents p_0, p write $p_{0,k} = (-\log e_k)^{p_0}$ and $p_k = (-\log e_k)^p$. Here $\mathrm{Box}_k = \{\sup_\Upsilon |\zeta(\Upsilon)| \le p_{0,k}\}$; its restriction to Box_k has probability $\exp(-\ell^3\delta\varepsilon_k^0)$, with $0 \le \delta\varepsilon_k^0 \le 12\exp(-p_{0,k}^2/2)$. The symbols C and $\mathscr{C}$ always denote, respectively, this real coordinate map and the external connection.

The common constant c_γ bounds both

$$\sup_\Upsilon \sum_{\Upsilon'} \exp[-\gamma_w d(\Upsilon, \Upsilon')], \qquad \sup_b \sum_{b'} \exp[-\gamma_w d(b, b')],$$

where the second sum is over unit bonds. The constants C_w and γ_w dominate the photon bounds and the corresponding fermion bounds as stipulated in the primitive input in section 3.

3. The primitive and nonlinear-step inputs

We use the photon and fermion constructions of [MB; FB], the extraction and reconstruction maps of [NC], and the nonlinear maps of [NL]. We recall their hypotheses and the conclusions needed here. The finite expressions used in the comparison are recalled in appendix A.

For an active step put $e = e_k$, $e_{k+1} = \sqrt{L}e_k$ and $0 < \mu \le L^{-1}$. Both charges lie below the common ceiling of the primitive estimates. The incoming and outgoing sides are admissible: $\ell/(LM)$ and $\ell/(L^2M)$ are odd integers at least 13. Assume $0 < \rho < 1$, odd $L \ge 4^8$, $M = L^m \ge L^2$, $M > 4L$, $0 < \varepsilon \le \bar{\varepsilon} = 1/200$, and $1 - \varepsilon < \alpha < 1$. The condition $\ell \ge Mr_k$ is imposed at every active step.

The constants in the block estimates are

$$c_X = 54, \qquad \kappa_0 = 54\log 72, \qquad K_{\mathrm{pol}} = \exp(\kappa_0)/36, \qquad c_{\mathrm{rb}} = c_X K_{\mathrm{pol}}, \qquad c_{\mathrm{fl}} = 61c_{\mathrm{rb}},$$
$$c_C = 3L^{9/2}, \qquad b = 1, \qquad b_L = b/L, \qquad b_- = 1 - L^{-1},$$
$$C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)],$$
$$c_S = 10^5 c_X^5 K_{\mathrm{pol}}^5 \exp(6\kappa), \qquad \kappa_w = \kappa + \kappa_0 + 3.$$

The constants $C_w, \gamma_w, \alpha_0, C_B$ and c_γ are those of the primitive families; $c_\gamma, C_w \ge 1$. Here c_γ bounds the exponential row sums on the photon coordinate and unit-bond index sets. The determinant constant is

$$C_d = 16c_X C_w \max\{\tfrac{\pi}{2} b\exp(1)(1 + 2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}.$$

For related tori C_w always means the common constant after projection, $64K_{\mathrm{pol}}C_w^{(1)}$.

Primitive hypotheses. Use the full hypotheses and conclusions of [MB, Theorems A–B and Proposition 5.1] and [FB, Theorem A and Propositions 4.1–4.3]. In particular the parameter families are defined on $|s_{\mathrm{pc}}| \le M^{\alpha_0}$, with common analytic kernel and short Hölder bounds, exact full-strength values, component restriction, the row-vanishing properties stated for each family, and local read bonds. Write $s = (s', s'')$, where $s' = (s'_M, s'_{LM})$ consists of the photon reconstruction and square-root parameters, of kinds π_M, π_{LM} in [MB, (37)]. The fermion parameters s'' are indexed by the partition

π_k into M-cubes, as in [FB, §4.1]. For a photon site c, put $K_c = \{c' : |c' - c|_\infty \le 2L\}$; the row of H_k at (c, g) vanishes unless every M-cube meeting K_c is active. Their lattice, gauge, transpose and reality identities are retained. At zero use the diagonal families and the full chamber covariance. The local resolvent bound then retains its factor $(1 + y)^{-1}$. The top fermion estimates are taken with polymer decay $\kappa_w + \kappa_0$; the projected estimates have decay κ_w.

The finite inverse and walk conditions of the fermion construction and the photon weakening condition

$$0 < \alpha_0 \le \tfrac{1}{4}, \qquad 2\alpha_0 n_P \log M \le \gamma_P M, \qquad n_P = 402$$

are assumed. Exact photon intertwining under odd covers is [MB, (46)]. The comparison below proves intertwining of the exact fermion formulas directly.

Extraction, reconstruction and determinant estimates. Use [NC, Theorems A–B and Appendix B], including the fine inverses at both signs of the total connection required by the reconstruction theorem. Use the weighted integration convention of [FB, Theorem B and Proposition 5.1] and the determinant estimates of [FB, Propositions 6.1–6.2]. The extraction maps are linear, real on real activities, and satisfy

$$|m(\mathscr{E})| \le c_{\mathfrak{fl}} e_k^{1/2} \|\mathscr{E}\|_k^x, \qquad |\varepsilon_{\mathrm{v}}(\mathscr{E})| \le c_{\mathfrak{fl}} \|\mathscr{E}\|_k^x.$$

The normalized linear term and determinant term satisfy

$$\|\mathcal{L}(R\mathscr{E})^{\mathrm{nat}}\|_{k+1}^x \le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon} \|\mathscr{E}\|_k^x, \qquad \|\mathcal{L}\delta E_k^{\det}\|_{k+1}^x \le e_{k+1}^{1/4-2\varepsilon}.$$

The scaled domains obey

$$\mathscr{A}_L \in L^{-3/4} \widetilde{R}_k \subset \widetilde{R}_k^-, \qquad \mathscr{C}_L \in L^{-3/4} \mathfrak{X}_k.$$

The complete real activity spaces, all these maps and their norms commute with restriction to the external slot $\mathscr{C} = 0$; this restriction does not increase the norm.

Numerical conditions for the nonlinear step. The hypotheses of [NL, Theorem A] include

$$L^{1/8-3\bar{\varepsilon}/2} \ge 2400, \qquad e^{\varepsilon} \le \tfrac{1}{4}, \qquad 160 L^{1+3\varepsilon/2} M e^{\varepsilon} \le 1,$$
$$9ML^{\varepsilon} e^{2\varepsilon} \le 1, \qquad 90 c_X L M e^{\varepsilon} \le 1, \qquad \kappa \ge 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \ge \kappa,$$

and

$$C_w \ge 4 \exp(2\gamma_w) M^{27\alpha_0}, \qquad M^{\alpha_0} - 1 \ge \exp((2L+1)^3 \kappa + \log(6 c_X)).$$

For $v = -\log e$, $p_{0,k} = v^{p_0}$ and $p_k = v^p$, impose

$$p_{0,k} \le e^{-\varepsilon/4}, \qquad 12 \exp(-p_{0,k}^2/2) \le e^7, \qquad 2e^{5\varepsilon/2} \le 1, \qquad c_{\mathfrak{fl}} e^{\varepsilon} \le 1, \qquad 3Me^{\varepsilon} \le 1,$$
$$2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \le 1, \qquad 16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \le 1, \qquad 96 M^3 e^{1/4} \le 1,$$
$$32 C_w C_T e^{3\varepsilon/2} \le 1, \qquad 88 M^3 e^{1/4-6\varepsilon} \le 1, \qquad 112 C_w^2 c_\gamma e^{2\varepsilon} \le 3,$$
$$c_S e^{1/4-5\varepsilon} \le 1, \qquad c_S L^3 e^{\varepsilon} \le 1, \qquad 8 c_{\mathrm{rb}} L^3 C_d M^3 e^{2\varepsilon} \le 1,$$
$$8 C_B p_k \le e^{-3/4+3\varepsilon}, \qquad 8 e^{3/4-7\varepsilon/2} \le 1, \qquad L^{3/4} \ge 8/3, \qquad v \ge 9, \qquad 8M(v/4+4) e^{\varepsilon} \le 1,$$
$$e^{1/4} + e^{\varepsilon} < 1, \qquad 4e^{1/4+2\varepsilon} + 2e^{3\varepsilon} \le 1.$$

These conditions are imposed at each active scale. The additional conditions used for the stopping arithmetic and the finite-sequence contraction are stated with the physical scales.

The imported nonlinear conclusions. The displayed finite contour definitions use

$$r_1 = \tfrac{1}{2} e^{-3/4+9\varepsilon/2}, \qquad r_2 = e^{-3/4+7\varepsilon}, \qquad r_4 = e^{-1/4+3\varepsilon}.$$

For a bond field Z put $|Z|_* = \max\{\|Z\|_\infty, \|\partial Z\|_\infty, \|\delta_\alpha \partial Z\|_\infty\}$. By [NL, Proposition 5.1], $2 \le r_2, r_4 \le r_1$ and

$$|Z(s')|_* \le \tfrac{1}{2} e^{-\varepsilon/2}, \qquad A + tZ(s') \in \widetilde{R}_k, \qquad A + \mathscr{C} + tZ(s') \in \widetilde{\mathcal{G}}_k \quad (|t| \le r_1).$$

The finite activities have the local ordinary, pair and fluctuation supports and read bonds stated in [NL, Proposition 5.3]. Their connected union expansion converges absolutely in the weighted norm and its exponential is the original finite fluctuation integral, by [NL, Proposition 6.1].

For every complex datum in $\mathrm{Ball}_k^x(4)$, the resulting correction satisfies

$$\|E_k^*(m,\mathscr{E})\|_{k+1}^x \le e_k^{1/4-6\varepsilon}.$$

It is norm analytic on affine complex lines in the interior of the allowed data ball, continuous on their closed allowed sets, and real for real data. For two data in $\mathrm{Ball}_k^x(1)$,

$$\|E_k^*(m,\mathscr{E}) - E_k^*(\widetilde{m},\widetilde{\mathscr{E}})\|_{k+1}^x \le e_k^{1/4-6\varepsilon}\left(\frac{|m-\widetilde{m}|}{w_k^{\mathrm{m}}} + \frac{\|\mathscr{E}-\widetilde{\mathscr{E}}\|_k^x}{w_k^{\mathrm{E}}}\right).$$

At $k=0$ the activity is fixed to zero and the second quotient is absent. These are [NL, Proposition 8.1]. The full mass, vacuum, activity and external-scalar recursions reproduce the exact density and its unnormalized photon-box marginal by [NL, Theorem A]. In particular $0 \le \delta\varepsilon_k^0 \le e_k^7$. All these conclusions commute with restriction to $\mathscr{C}=0$.

The exact external scalar update, with $\mathfrak{P}_0 = 0$, is

$$\mathfrak{P}_{k+1}(Y,\mathscr{C}') = \sum_{\bar{X}^L = LY} [\mathfrak{P}_k(X,\mathscr{C}'_L) + E_k^{\det}(X,\mathscr{C}'_L)]. \tag{14}$$

The full activity update is

$$\mathscr{E}^+ = \mathcal{L}(R\mathscr{E})^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m,\mathscr{E}), \qquad \delta E_k^{\det}(X;A,\mathscr{C}) = E_k^d(X,A+\mathscr{C}) - E_k^d(X,\mathscr{C}).$$

With no external argument the determinant difference is $E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0)$. At the first step the incoming activity vanishes. There are $192\ell_0^3$ incoming chamber pairs of measure $1/48$ and $4\ell_1^3$ outgoing pairs. The fluctuation covariance is 48δ on the chamber modes, and the local vertex is (72).

4. Projection of the primitive systems

We construct the primitive systems on one large torus and project them to smaller tori through odd covering maps. Projection preserves the estimates with a fixed loss in the common constant and one loss in polymer decay. The comparison of local steps then uses supported rows near a common lift, where the signed image sum has only one surviving term. This is the construction of [QED, §§15.1–15.3].

4.1. **Odd covers and projected kernels.** Fix $k \ge 0$, $0 < \mu \le 1$, $0 < e_k \le \bar{e}_0$, and admissible sides $\ell_2 = J_c\ell_1$, where J_c is odd. Write $T_i = \eta_k\mathbb{Z}^3/\ell_i\mathbb{Z}^3$ and $p : T_2 \to T_1$ for the cover. Let pr_i denote the covering-lattice projections. Pullback p^* is inclusion of functions on the covering index sets. They have character $\chi(w) = 1$ for photon fields, coordinates and parameters, and $\chi(w) = (-1)^{w_1+w_2+w_3}$ for spinors and pair fields. Oddness gives $\chi(J_c w) = \chi(w)$, so a field on T_1 is a field on T_2 with the same character. The image is

$$\mathrm{Per} = \{f : f(t+\ell_1 w) = \chi(w)f(t),\ w \in \mathbb{Z}^3\}.$$

Translations move the photon site and preserve the chamber label. In particular pullback commutes with J, J^*, complex conjugation and constant spin matrices.

The index weight is 1 on unit sites, unit bonds and photon coordinate indices, η_k^3 on fine photon sites and bonds, and $\eta_k^3/48$ on chamber sites. For fields with a common index set and character,

$$\|p^*f\|_\infty = \|f\|_\infty, \qquad \langle p^*f', p^*f\rangle_{T_2} = J_c^3\langle f', f\rangle_{T_1}. \tag{15}$$

Indeed a fundamental set for T_2 is the union of $\mathrm{Fnd}_1 + \ell_1 w$, over representatives $w \in (\mathbb{Z}/J_c\mathbb{Z})^3$. The two characters cancel in each paired summand, and the weight is the same on both tori. At zero the chamber weight remains $1/48$.

Pullback maps the parameter polydisc, including its interior, into the corresponding larger polydisc, since $(p^*s)_{\mathrm{pc}} = s_{p\,\mathrm{pc}}$. It also maps $\widetilde{\mathcal{G}}_k(T_1)$ into $\widetilde{\mathcal{G}}_k(T_2)$. To check the latter assertion, write the domain decomposition $A = A_0 + A_1$. Pullback preserves reality of A_0 and the sup norm of A_1. Each five-cube neighborhood has a common nonwrapping lift of side $11M < \ell_1$. The real local gauge representative of A_0 on its image in T_1 pulls back on this lift to the corresponding neighborhood in T_2, with its same sup bound and gradient identity. This is precisely the local real-gauge condition in the inverse domain.

Every exact finite fermion operator used below intertwines with pullback. For the chamber Dirac operator this follows term by term from its four neighbors and bond phases. For $\mathcal{Q}_k = Q_k J^*$ and its reverse it follows from the same block sums and the same six paths on the cover; J^* averages the same 48 labels. The penalty and the finite matrices multiplying the resolvent have the same finite block formulas. Sums and products preserve intertwining. If $O_2 p^* = p^* O_1$ and both operators are invertible, then $O_2 p^* O_1^{-1} = p^*$ and hence $O_2^{-1} p^* = p^* O_1^{-1}$. The domain preservation just proved makes the already established inverse bounds applicable on both tori. It follows, from the formulas in Section 4 of [FB], that the fine resolvents, $H_k, \bar{H}_k, D_k, M_k, \Gamma_k$ intertwine at the same connection:

$$O_2(p^*A)p^* = p^* O_1(A). \tag{16}$$

At zero use the entire chamber Dirac stencil and the inverse of F_y, on all chamber components. The exact photon operators have the corresponding identity (46) of [MB].

Let a primitive system on T_2 be given. For a parameter family $O_2(s, A)$ put $\widetilde{O}(s, A) = O_2(p^*s, p^*A)$. For a polymer family put

$$\widetilde{O}(X, A) = \sum_{X' \in \mathcal{P}_k(T_2) : pX' = X} O_2(X', p^*A).$$

This is applied to M_k for $k \geq 1$ and to $S_{k,y}$ at every level. A projected polymer is connected: adjacent M-cubes have equal or adjacent images. Moreover

$$d_M(pX') \leq d_M(X'). \tag{17}$$

A projected net meets all image cubes. Projecting its chosen finite segment representation preserves the lengths of the segments; taking the infimum gives (17). There are only finitely many polymers on either torus.

Proposition 4.1. If the system has translation covariance under $\ell_1\mathbb{Z}^3$, its projection is well defined by $p^*O_1 = \widetilde{O}p^*$ and has kernel

$$O_1(\cdot)(t, t') = \sum_{w \in (\mathbb{Z}/J_c\mathbb{Z})^3} \chi(w)\widetilde{O}(\cdot)(t, t' + \ell_1 w). \tag{18}$$

This formula uses the same index weights on both tori. The signed summand depends only on the class of w. Transposition of an intertwining operator also intertwines:

$$O_2^T p^* = p^* O_1^T. \tag{19}$$

■

Proof. The pulled-back parameters and connection are invariant under ℓ_1 translations. Parameter-family covariance therefore says that $\widetilde{O}$ commutes with those translations. For polymer families, translation permutes the polymers X' with $pX' = X$, so the same commutation holds after their finite sum. An operator commuting with these translations maps the character eigenspace Per into itself. Since p^* is injective onto that space, this defines exactly one projected operator.

Sum its action over the fundamental sets used in (15) and insert $f(t' + \ell_1 w) = \chi(w)f(t')$. The coefficient of $\nu(t')f(t')$ is (18). Replacing w by $w + J_c v$ multiplies the character by $\chi(v)$ and the large-torus kernel by the same character, so the product is unchanged. This proves the asserted kernel formula, including the weights $\eta_k^3/48$ or $1/48$ on chamber columns.

Translations have transpose $T_v^T = T_{-v}$ for the displayed bilinear pairings. Hence O_2^T also commutes with ℓ_1 translations and maps Per into itself. Write $O_2^T p^* f' = p^* g$. For every source field f, intertwining and (15) give

$$J_c^3 \langle g, f\rangle_{T_1} = \langle O_2^T p^* f', p^* f\rangle_{T_2} = J_c^3 \langle f', O_1 f\rangle_{T_1}.$$

Nondegeneracy gives $g = O_1^T f'$. □

4.2. **The uniform projection estimate.** The supports of primitive row and column indices have sup diameter at most L. Assume $\gamma_w M \geq 1$. Then

$$\sum_{w \in (\mathbb{Z}/J_c\mathbb{Z})^3} \exp\big(-\gamma_w d_2(t, t' + \ell_1 w)\big) \leq 64 \exp\big(-\gamma_w d_1(t, t')\big). \tag{20}$$

To prove this, translate the column lift by an ℓ_1 period so that the distance of the two support sets equals $d_1 \leq \ell_1/2$. For each class of w, choose a representative w' which realizes its distance on T_2. Distinct classes have distinct representatives. Their distances are at least d_1; for $|w'|_\infty = j \geq 2$ they are at least

$$\ell_1 j - d_1 - 2L \geq d_1 + (\ell_1 - 2L)(j-1).$$

This uses the support diameters and $d_1 \leq \ell_1/2$. Since $\ell_1 \geq 13LM$, we have $\gamma_w(\ell_1 - 2L) \geq 6$. There are 27 lattice vectors of norm at most 1 and at most $26j^2$ of norm j. The sum is therefore bounded by

$$\exp(-\gamma_w d_1)\left(27 + \sum_{j \geq 2} 26 j^2 \exp(-6(j-1))\right).$$

The latter series starts below 0.269 and its successive ratios are at most $(9/4)\exp(-6) < 1/100$. Thus the bracket is below $27.3 < 64$, proving (20).

There is a second, polymer summation estimate. For a unit or chamber row index t, let $\square_t$ be its M-cube on T_2. By (17) and counting,

$$\sum_{\substack{pX' = X \\ X' \supseteq \square_t}} \exp\big(-(\kappa_w + \kappa_0) d_M(X')\big) \leq K_{\mathrm{pol}} \exp\big(-\kappa_w d_M(X)\big). \tag{21}$$

Indeed the factor with rate κ_w is at most the factor on the right; the remaining sum is bounded by $\sum_{X' \supseteq \square_t} \exp(-\kappa_0 d_M(X')) \leq K_{\mathrm{pol}}$. The additional top-system decay κ_0 pays for the polymer image sum.

Proposition 4.2. Suppose the system on T_2 has the primitive properties with parameters

$$(\alpha_0, \gamma_w, \kappa_w + \kappa_0; C_w^{(1)}), \qquad C_w^{(1)} \geq 4 \exp(2\gamma_w) M^{27\alpha_0}.$$

Its projection has the corresponding properties on T_1 with

$$C_w = 64 K_{\mathrm{pol}} C_w^{(1)} \tag{22}$$

and parameters $(\alpha_0, \gamma_w, \kappa_w; C_w)$. In particular it satisfies the primitive hypotheses of Theorem A of [NL]. At zero the resolvent estimate retains $(1+y)^{-1}$. ■

Proof. We verify the identities, estimates and local structure needed for the step. At full strength the exact intertwining identities and the primitive sum identities on T_2 give

$$H_{k,1}^{\mathrm{ph}}(1) = H_{k,1}^{\mathrm{ph}}, \quad C_{k,1}^{1/2}(1) = C_{k,1}^{1/2}, \quad H_{k,1}(1, A) = H_{k,1}(A), \quad \Gamma_{k,1}(1, A) = \Gamma_{k,1}(A).$$

For the two polymer families, sum first over all X: each X' occurs once, under $X = pX'$. Equation (16) then identifies the sum with the exact operator on T_1. All projected entries are finite sums of original entries composed with the linear pullback. Continuity on the closed parameter polydisc, holomorphy in its interior and in the connection for any fixed parameter, and Borel dependence on the resolvent parameter follow from these finite sums.

For the photon reconstruction, its first and Hölder derivative kernels have the same image-sum formula: the difference acts in the row variable, and short lifted differences agree on the two tori. Equation (20) bounds each by $64C_w^{(1)}\exp(-\gamma_w d_1)$. It gives the same bound for the square-root kernel and for Γ_k. The H_k sup and short-pair Hölder bounds instead follow by the isometric pullback in (15); their constant is $C_w^{(1)}$. A short pair and its six paths lift identically. An intracluster quotient is $\eta_k^{-\alpha}(f(c,g')-f(c,g))$ on each torus.

For $M_k(X,A)$, only X' containing $\square_t$ contribute to its row, by original polymer support. Combine (18), (20) and (21) to get

$$|M_{k,1}(X,A)(t,t')|_c \le C_w \exp(-\gamma_w d_1(t,t') - \kappa_w d_M(X)).$$

For a positive-level fine resolvent, the pullback of a unit-block indicator is the disjoint sum of its J_c^3 translated large-torus block indicators. The same two estimates bound its block operator norm. At zero use its chamber kernel directly: the original factor $(1+y)^{-1}$ is common to all terms of both image sums and remains on the right. All column weights are unchanged, so no further factor 48 appears.

For component locality, use shared-site linked components for the photon parameter cubes and face-connected components for the fermion partition M-cubes. An index whose support is contained in its active union belongs to exactly one such component. Fermion site supports are singletons, each in one partition cube. For the photon kinds, the geometric argument is as follows. For a bond of length $a \in \{\eta_k, 1\}$, cubes of radius M meeting its support either share an endpoint or would have centers differing in that direction by $2M+a$. The latter is impossible since the centers differ by a multiple of M and $0 < a < M$. For a coordinate index, its support has diameter at most L. The centers of two cubes of radius LM meeting it differ coordinatewise by less than $3LM$, and are multiples of LM; choosing the coordinate midpoint when the difference is $2LM$, and either center when it is LM, exhibits a lattice site in both cubes. Thus all cubes meeting the support are linked. The same argument applies on either torus by lifting the support and these cubes into a nonwrapping neighborhood.

Pull back the active parameter cubes on T_1. A component on T_2 maps into one component on T_1, since the covering map preserves shared sites for photon cubes and common faces for fermion partition cubes. For a row in a small-torus component, the unique large-torus component containing its lift maps into that component. Original component restriction confines all surviving columns there. Changing the parameters and connection outside the small component leaves these original rows unchanged; the signed image sum therefore preserves their row dependence. The identical argument applied to each lifted column preserves column component restriction. Each row-vanishing property stated for an original family passes to its projection. In particular, if a cube meeting K_c is missing, its unique lift meeting the lifted K_c is missing too, so the H_k row vanishes. For $k \ge 1$, the parameter-independent term $\mathcal{B}_{k,0}$ in Γ_k is retained; it preserves every union of partition M-cubes [FB, Proposition 4.3 and its proof]. For a projected polymer family a nonzero term has both original indices in X', so both projected indices lie in X. Equality of connections on bonds meeting X gives equality on bonds meeting every such X', preserving the polymer's local dependence.

Gauge covariance passes through pullback because the local gradients, restriction to unit sites and phases commute with it. Lattice covariance follows by lifting a symmetry (S,a) with one representative $a \in LM\mathbb{Z}^3$ to the same affine map on both tori; it permutes the image polymers. A change of representative multiplies source and target spinors by the same sign, which cancels in the operator relation. Transposition passes by (19); the constant C commutes with pullback. Complex conjugation and $\varkappa$ also commute with it. These facts prove the charge and reality relations for the projected primitives.

At zero, the photon and fermion diagonal factors are unchanged by projection. Cubes containing a given bond, and cubes meeting K_c, have unique corresponding lifts, so their parameter products are the same on both tori. The fermion factor is the same for all 48 labels at c. Their constants are

bounded by $4\exp(2\gamma_w)M^{27\alpha_0} \le C_w^{(1)}$. The remaining zero-level properties are those just proved for the square root, chamber covariance and local resolvent. This completes the system verification with the common constant (22). □

4.3. **A common top-torus system.** On a chosen top torus use the photon families (37) of [MB] and the local fermion families (57) of [FB]. In the fermion walk construction take the polymer decay rate $\kappa' = \kappa_w + \kappa_0$. The formulas do not depend on this rate; it specifies the parameter restrictions and the constants in their estimates. We write $\gamma^F = \gamma$ for the fermion decay rate in (31) of [FB]. Define

$$\alpha_0 = 1/54, \qquad \gamma_w = \min(\gamma^F, \gamma^P), \quad \gamma^P = \gamma_P/2,$$
$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w)M^{27\alpha_0}\}, \qquad C_w = 64 K_{\mathrm{pol}} C_w^{(1)}. \tag{23}$$

Here C^F, γ^F are the constants of the fermion construction at the indicated polymer rate. The fixed M satisfies both its fermion condition and (38) of [MB]; the charge satisfies the inverse and walk conditions of those constructions.

These families form a system with $(\alpha_0, \gamma_w, \kappa_w + \kappa_0; C_w^{(1)})$. This follows by matching their proved properties. The photon kernel multipliers are polynomials in their parameter cubes; at full strength the partition of unity and the products give the exact kernels. Their path products confine rows and columns to their active link-components. The photon derivative and square-root bounds have constants C^P and rate γ^P. The fermion families are the convergent weakened walks and the localized block expressions, with their proved support, covariance, analyticity and Hölder estimates. Replacing their rates γ^F, γ^P by the smaller γ_w and their constants by the larger $C_w^{(1)}$ preserves every inequality. The row and column restriction identities contain no decay constant and are unchanged.

At level zero the two diagonal families are used. The photon derivative bound has constant $4\exp(2\gamma_w)M^{27\alpha_0}$, and the fermion diagonal factor is bounded by $M^{8\alpha_0}$, below that constant. There is no primitive short pair with $0 < d_X < 1$ at $\eta_0 = 1$. The covariance and local resolvent are chamber kernels with measure $1/48$, and the resolvent bound includes $(1+y)^{-1}$. A unit block has total chamber measure 1, since $(\eta_k^3/48)(48\eta_k^{-3}) = 1$. Consequently the positive-level block estimates and zero-level chamber estimates use precisely the common constants (23). Projecting this system by proposition 4.2 gives a system with these primitive bounds and the same C_w on every odd smaller torus. The top system itself has these weaker bounds as well. None of these constants depends on the level, side, mass, charge or harmonic background.

5. Agreement of local steps

5.1. **Common lifts and local algebras.** For the rest of the comparison assume

$$\ell_1 > (31L + 16)M. \tag{24}$$

Put $d_U = 2L + 1$, $c_l = 2L$. For a finite nonempty union $\widetilde{X}$ of covering M-cubes, let $\mathrm{box}(\widetilde{X})$ be its smallest rectangular union of cubes and let $\mathrm{diam}(\widetilde{X})$ be that box's largest side. Define

$$N(\widetilde{X}) = \mathrm{box}(\widetilde{X})^{\sim 6d_U}.$$

We call $\widetilde{X}$ admissible when $\mathrm{diam}(\widetilde{X}) \le \ell_1 - 14 d_U M$; a polymer is flat when it is the image of a connected admissible $\widetilde{X}$. These are the sets on which the two torus algebras will be identified.

Proposition 5.1. For admissible $\widetilde{X}$, projection is injective on $N(\widetilde{X})$ and on every supported index contained there. An index of either torus supported in the image of a subset of $N(\widetilde{X})$ has exactly one lift contained in that subset. Pairs have their shortest lifts, with the same d_X and six paths. Local incidences of total support diameter below $2d_U M$ are preserved in both directions.

There is a norm-preserving algebra isomorphism $\iota_{\widetilde{X}}$ between the two subalgebras generated by ordinary fields over $N(\widetilde{X})$, pair fields with both endpoints there, and their unit fluctuation and

auxiliary fields there. It identifies covering generator indices. At zero it includes the chamber W fields and no auxiliary Ψ. ■

Proof. Every side of $N(\widetilde{X})$ is at most $\ell_1 - 2d_U M$. Since faces contain no fine lattice sites, any two of its sites satisfy

$$|u - u'|_\infty \le \ell_1 - 2d_U M - \eta_k.$$

For $w \neq 0$ this implies

$$|u - u' - \ell_i w|_\infty \ge 2d_U M + \eta_k. \tag{25}$$

It proves disjointness of $N(\widetilde{X})$ from its nonzero ℓ_1 translates, injectivity, and uniqueness of any representative with distance below $2d_U M$.

To lift an index supported in the projected subset, translate one of its lifts so that one support point lies in that subset. Every remaining support point is congruent to a point of the subset and is within $2LM < 2d_U M$ of the first. Equation (25) forces the congruence translation for each point to be zero. Thus the whole index lies in the subset; injectivity gives uniqueness. In particular the shortest pair has its endpoints there, distance at most M, and its same covering paths.

For an incidence relation, lift a tuple realizing the projected relation and translate its first index to the prescribed lift. Each other lifted index differs from the prescribed one by an ℓ_i period. Its support is at distance below $2d_U M$ from the first one. Equation (25) again forces the period to vanish. This proves the reverse incidence implication; the forward one is projection. The relations used below meet this range condition: nearest neighbors and chamber neighbors, short pairs and their paths, block and cube membership, common faces, linked parameter cubes, cube meeting, and coordinate support. The largest diameter is $4LM < 2d_U M$, for two linked photon parameter cubes. They are all invariant under $LM\mathbb{Z}^3$ translations.

No two covering generator indices in $N(\widetilde{X})$ have the same period class on either torus. Relative to any fundamental generator system their images are therefore independent generators, up to their antiperiodic signs. Identifying the same covering indices extends uniquely to a free Grassmann algebra isomorphism. It preserves bar indices and degrees and sends basis monomials to basis monomials, up to the same coefficient signs. All generator weights and index measures are common, so it preserves norms. The maps for nested admissible sets restrict to one another. □

A local activity on $X_i = \mathrm{pr}_i \widetilde{X}$ belongs to this identified algebra. Ordinary sites lift to $\widetilde{X}$; an admissible pair lifts to $\widetilde{X}^{\sim 1}$. For its first kind the two sites and paths lie in $\widetilde{X}$. For its second kind the pair is $(x, x^{(3)})$ with an endpoint over $\widetilde{X}$. Each assertion follows from the incidence comparison. Two local families F_1, F_2 agree on flat polymers when

$$\iota_{\widetilde{X}} F_1(X_1; A_1, \mathscr{C}_1) = F_2(X_2; A_2, \mathscr{C}_2) \tag{26}$$

whenever the two admissible connection pairs have equal lifts on $N(\widetilde{X})$. It suffices to check $A_2 = p^* A_1$, $\mathscr{C}_2 = p^* \mathscr{C}_1$: for other pairs with equal lifts, locality of F_2 identifies them with these pullbacks on all bonds meeting $\widetilde{X}$. Thus once agreement is proved, equality holds already under equality on those read bonds.

Theorem B. Suppose the two sides at levels k and $k+1$ are admissible and satisfy $\ell_1/L > (31L + 16)M$. Use the projected and top systems constructed above, with the hypotheses of Theorem A of [NL] on both tori. Let the real mass m be common with $|m| \le w_k^{\mathrm{m}}$, and let the incoming real activities satisfy $\|\mathscr{E}_i\|_k^x \le w_k^{\mathrm{E}}$ and agree on flat polymers, with $\mathscr{E}_i = 0$ at zero. For the external scalar conclusion, assume the incoming scalar families agree on flat polymers as well. Then their extracted mass and vacuum contributions are equal. Their outgoing activities agree on flat polymers at level $k+1$. The external scalar recursion (14) preserves agreement of scalar families on such polymers. All these assertions hold for the activities without an external slot as well. ■□

The proof is given in section 5.6, after comparison of the local operations.

Every small polymer is flat. The net diameter bound gives $\operatorname{diam}(\widetilde{X}) \le M(d_M(\widetilde{X})+3) \le M(L+3)$ for a small connected lift. By (24), $M(L+3)+14d_UM < \ell_1$. The nonwrapping lift proposition then identifies small torus polymers containing a chosen cube with small covering polymers containing its chosen lift. This identification is the same on both tori. Conversely, if an admissible connected lift projects to a small polymer, the small nonwrapping lift is its period translate: start at one common cube and propagate equality along adjacent cubes using the incidence result. Hence smallness, tree length and volume coincide for its two images.

Subpolymers and parameter cubes contained in an admissible $\widetilde{Y}$ correspond bijectively on the two tori. Common faces and links are preserved, so incompatibility is preserved. The definitions by cube meeting give

$$P_0(\operatorname{pr}_i \widetilde{X}) = \operatorname{pr}_i P_0(\widetilde{X}), \qquad \operatorname{pr}_i Y'(\widetilde{X},\widetilde{P}) = Y'(\operatorname{pr}_i \widetilde{X}, \operatorname{pr}_i \widetilde{P}).$$

When $\widetilde{X}$ and all cubes of $\widetilde{P}$ are contained in $\widetilde{Y}$, the resulting Y' is contained there too, since the M-cubes partition sites. Reblocking satisfies $\overline{\widetilde{X}}^{L} \subset \widetilde{Y}^{\sim(L-1)}$ and commutes with projection. These larger sets still lie in $N(\widetilde{Y})$, so their incidences have the same unique lifts.

5.2. **Supported rows near a flat set.** The preceding identification applies to the weakened primitives because setting parameters to zero separates their components. We first record the component argument. Let s vanish at every parameter cube not contained in a union Z of M-cubes. If Z_1 is a face-connected component of Z and $\square(t)^{\sim 2L} \subset Z_1$, then

$$\begin{aligned} O(s,A)(t,t') &= 0 \quad \text{unless } t' \text{ is contained in } Z_1, \\ O(s,A)(t,\cdot) &= O(s\mathbf{1}_{Z_1}, A')(t,\cdot). \end{aligned} \tag{27}$$

Here O is a parameter primitive, and A' is any connection in the primitive's domain equal to A on bonds meeting Z_1; the photon primitives have no connection argument. The set $\square(t)$ is the union of the M-cubes meeting the support of t.

To prove (27), choose a parameter cube containing that support inside $\square(t)^{\sim 2L}$. For a chamber site or a unit site it is its M-cube. For a fine bond it is the parameter cube of radius M about the M-cube center of one end: both ends lie in it, and it is contained in $\square(t)^{\sim 1}$. For a photon coordinate index choose the parameter cube of radius LM about the LM-cube center of one point of its support. The support diameter is at most L and $M \ge 2$, so this cube contains the entire support. If z_c is the M-cube center of that point and z'' is the M-cube center of any site in the parameter cube, then

$$|z'' - z_c|_\infty < (3L/2+1)M \le (2L+1)M.$$

These centers differ by a multiple of M, hence the cube is contained in $\square(t)^{\sim 2L}$.

Let $\mathscr{Y}$ be all parameter cubes of the chosen kind contained in Z, and let $\mathscr{Y}_1$ be the component containing the cube just chosen, using shared-site links for photon cubes and face adjacency for fermion partition cubes. The M-cubes meeting $\bigcup \mathscr{Y}_1$ form a face-connected polymer contained in Z. It meets $\square(t) \subset Z_1$, and therefore is contained in the component Z_1. The component restriction property of the weakened primitives, proved in the photon and fermion constructions, now confines the row to $\bigcup \mathscr{Y}_1$. Replacing s by $s\mathbf{1}_{Z_1}$ leaves all its parameters there unchanged; replacing A by A' leaves all its read bonds unchanged. This proves (27).

Proposition 5.2. Let $\widetilde{Y}$ be admissible, $Y_i = \operatorname{pr}_i \widetilde{Y}$, and $0 \le j \le 5d_U$. Let s_1 vanish outside Y_1. Define s_2 by giving the corresponding parameter cubes contained in $\widetilde{Y}$ the same values and setting all other entries to zero. Suppose $A_i \in \widetilde{\mathcal{G}}_k(T_i)$ have equal lifts on the bonds contained in $\widetilde{Y}^{\sim(j+d_U)}$. For each parameter primitive and each covering row t contained in $\widetilde{Y}^{\sim j}$, the two rows, written with covering indices in $N(\widetilde{Y})$, coincide:

$$O_1(s_1,A_1)(t,t') = O_2(s_2,A_2)(t,t'). \tag{28}$$

Both rows have columns contained in $Y_i^{\sim(j+c_l)}$, and read the connection only on bonds contained in $Y_i^{\sim(j+d_U)}$.

For a connected nonempty $\widetilde{X} \subset \widetilde{Y}$ and connections with equal lifts on bonds meeting $\widetilde{X}$, the local polymer families also satisfy

$$O_1(\mathrm{pr}_1 \widetilde{X}, A_1)(t,t') = O_2(\mathrm{pr}_2 \widetilde{X}, A_2)(t,t') \tag{29}$$

at rows and columns over $\widetilde{X}$. Here $O = M_k$ for $k \geq 1$, or $O = S_{k,y}$ for any k and $y \geq 0$. ■

Proof. Put $\widetilde{B} = \widetilde{Y}^{\sim(j+c_l)}$. Its one-cube enlargement is $\widetilde{Y}^{\sim(j+d_U)} \subset N(\widetilde{Y})$. For the row t, the set $\square(t)$ is one M-cube or two face-adjacent M-cubes. This follows directly for site and bond indices; a coordinate support lies in one L-block or is the support of one bond. Thus $\square(t)^{\sim c_l}$ is a connected box contained in $\widetilde{B}$. Let $\widetilde{B}_1$ be the component of $\widetilde{B}$ containing this box.

Its image is a component of $\mathrm{pr}_1 \widetilde{B}$. On T_2 its image is a component both of $\mathrm{pr}_2 \widetilde{B}$ and of $p^{-1}(\mathrm{pr}_1 \widetilde{B})$. Indeed a cube adjacent to its image has a lift adjacent to a cube of $\widetilde{B}_1$. This lift lies in $\widetilde{B}^{\sim 1} \subset N(\widetilde{Y})$. If it belongs to a period translate of $\widetilde{B}$, separation (25) forces that translate to be zero. It then lies in $\widetilde{B}$ and in the same component. This proves maximality of the images as components.

Apply (27) on T_1. It confines the row to $\mathrm{pr}_1 \widetilde{B}_1$ and lets the connection be changed off the bonds meeting that component. Such bonds lift into $\widetilde{B}^{\sim 1}$, proving the claimed read set. On the parameter cubes contained in $\mathrm{pr}_2 \widetilde{B}_1$, the families $p^* s_1$ and s_2 agree. To see this even at cubes not contained in $\widetilde{Y}$, observe that each such cube in $N(\widetilde{Y})$ has a unique lift; its image cannot also be the image of a cube in $\widetilde{Y}$. Both entries are therefore zero. The connections $p^* A_1$ and A_2 agree on bonds meeting $\mathrm{pr}_2 \widetilde{B}_1$. Applying (27) twice on T_2 gives

$$O_2(p^* s_1, p^* A_1)(t,\cdot) = O_2(s_2, A_2)(t,\cdot),$$

with columns confined to $\mathrm{pr}_2 \widetilde{B}_1$.

In the signed image formula (18), a nonzero summand at $t' + \ell_1 w$ would have a lift in $\widetilde{B}_1$. Both this lift and t' belong to $N(\widetilde{Y})$; their difference is an ℓ_1 period. Separation forces $w = 0$ in $(\mathbb{Z}/J_c\mathbb{Z})^3$. That summand has character 1, proving (28).

For the polymer families, every polymer X' on T_2 with $pX' = \mathrm{pr}_1 \widetilde{X}$ is $\mathrm{pr}_2(\widetilde{X} + v)$ for one $v \in \ell_1\mathbb{Z}^3/\ell_2\mathbb{Z}^3$. Distinct classes give disjoint cube sets. To verify the assertion for an arbitrary X', lift two adjacent cubes as cubes of $\widetilde{X} + v_1$ and $\widetilde{X} + v_2$. Their adjacency places a translate of the second cube in $\widetilde{X}^{\sim 1} \subset N(\widetilde{Y})$. Separation forces $v_1 = v_2$ modulo $\ell_2\mathbb{Z}^3$. Connectedness gives one class for every cube of X', and projection onto the whole image gives equality with that entire copy.

The projected kernel is consequently a double sum over v and the column image w. The local polymer support requires both its row and column to be over $\mathrm{pr}_2(\widetilde{X} + v)$. The common row lift forces $v = 0$; the common column lift then forces $w = 0$. The remaining term is $O_2(\mathrm{pr}_2 \widetilde{X}, p^* A_1)(t,t')$. Locality on the bonds meeting this polymer permits $p^* A_1$ to be replaced by A_2.

The barred mean field satisfies $\bar{H}_k(s,A) = \mathsf{C}^{-1} H_k(s,-A)\mathsf{C}$, so its comparison follows at the connections $-A_i$. At zero the mean-field kernels are $48\delta_{x,x'}\mathbf{1}_4$ times their parameter products; the coordinate and chamber measures are the same on the two tori, and the chamber label is preserved. There is no M_0 family. The proof applies to Γ_0 and $S_{0,y}$ on the full chamber indices. □

5.3. **Finite rows and the local operations.** For finite operators, equality of rows follows directly from their covering formulas. Suppose a row has all its columns in $N(\widetilde{Y})$, and its connection and polymer membership data agree on the bonds and sites which that row reads. The two formulas are then the same finite sum on either torus. There are no repeated column classes by proposition 5.1. Writing the row as $\sum_{t'} r(t,t') f(t')$, its kernel is

$$O(t,t') = \nu(t')^{-1} r(t,t'). \tag{30}$$

Thus equality holds for the kernels as well, with the same column weight.

We need this observation at connections B_i having equal lifts on bonds meeting a connected $\widetilde{U} \subset \widetilde{Y}$. The photon coordinate map C reads no connection: a noncentral face bond is its own coordinate, and the central bond is the negative sum of the other face bonds and the interior straight-path contributions $n_c(b)/L$. All columns are supported in the one or two L-blocks of the row bond's ends. Hence its rows at bonds in $\widetilde{U}^{\sim(1+2d_U)}$ agree and have range at most $2L-1$.

The covariant difference row at an admissible pair is

$$d_X(x,x')^{-\alpha}\big(U_B(x,x')f(x') - f(x)\big).$$

Its only columns are x, x', and it reads exactly the six pair paths. These paths are contained in $\widetilde{U}$ for the first kind, or are the bond meeting $\widetilde{U}$ for the second kind. The range is M; inside a chamber cluster the transport is 1 and no bond is read. These rows therefore agree, also at $-B_i$.

For $k \geq 1$ the reconstruction rows $T^{(1)}, T^{(2)}$ and their barred versions are the two-piece formulas of the reconstruction estimate. At $y \in \widetilde{U}$ they have columns over Δ_y and at $x^{(3)}$, or at the pair $(x, x^{(3)})$ across its boundary. They read the bonds in Δ_y and the bonds (c, c_3) starting there. All meet $\widetilde{U}$. Their membership indicators are the same because

$$u \in \widetilde{U} + \ell_i\mathbb{Z}^3 \quad\Longleftrightarrow\quad u \in \widetilde{U} \qquad (u \in \widetilde{U}^{\sim 1}),$$

again by separation. Every site involved is at distance at most $(1+\eta_k)/2 < 1$ from y. This proves equality of both pieces, including the boundary term.

The finite rows of $Q_k J^*$ and $JQ_k^T(-B)$ have their columns and read bonds in a single Δ_y. The row of $B_{k,r}$, where $r \geq 0$ is the resolvent parameter, has its columns in one L-block and reads the paths from its center to its sites. These paths lie in that block. Their kernels are

$$\begin{aligned}(Q_k J^*)(y,x) &= \mathbf{1}[c \in \Delta_y]\varphi_B(y,c)\mathbf{1}_4,\\ (JQ_k^T(-B))(x,y) &= \mathbf{1}[c \in \Delta_y]\varphi_B(y,c)^{-1}\mathbf{1}_4,\\ B_{k,r} &= (b_k + \mathrm{i}\gamma_5 r)^{-1} I\\ &\quad + \big((b_k + b_L + \mathrm{i}\gamma_5 r)^{-1} - (b_k + \mathrm{i}\gamma_5 r)^{-1}\big)P_B.\end{aligned}$$

The projection P_B has kernel L^{-3} times the two block transports when its indices lie in the same L-block, and zero otherwise. The first two rows have range below 1, the third range at most $L-1$. At rows in $\widetilde{U}$ their columns and read bonds stay in its M-cube. They agree on both tori.

At zero, F_0 has only its diagonal, the four chamber neighbors, and its block-penalty columns. The hopping part of its kernel is

$$48\Big((\mu + \tfrac{\rho}{2}\sum_{i=0}^{3} w_i)\delta_{x,x'}\mathbf{1}_4 - \sum_{j=0}^{3}\delta_{x',x^{(j)}}M_j(x)U_B(x,x^{(j)})\Big).$$

The factor 48 is the inverse column weight. Only the fourth neighbor reads a bond meeting $\widetilde{U}$; the penalty reads paths inside the L-block of c. All columns lie over $\widetilde{U}^{\sim 1}$ and its range is at most $L-1$. Thus this entire row agrees at zero.

These equalities also identify the mass form with the covering sum:

$$\langle\bar\psi,\psi\rangle_{\mathrm{pr}_i\widetilde{\square}} = (\eta_k^3/48)\sum_{c\in\widetilde{\square},g,\beta}\bar\psi_\beta(c,g)\psi_\beta(c,g).$$

At zero the two halves of the first-step vertex have the common coefficient $1/96$, the same two boundary transports, and the same generators over the cube and at its outward pairs. Hence the first vertex also agrees under $\iota_{\widetilde{Y}}$.

The remaining local operations can be compared in this common free algebra. Equality of elements gives equality of their constant coefficients and of their antisymmetric quadratic kernels: expand in ordered covering monomials and use the common measure $\eta_k^3/48$. For a small polymer,

the adjustment $\Theta_{X,B}$ uses averages over shortest chains of M-cubes in X. Common faces, path membership and transports are identical on its lift, as are its volume and the coefficients d_X^α. Therefore

$$\iota_{\widetilde{X}}\Theta_{X_1,B_1} = \Theta_{X_2,B_2}\iota_{\widetilde{X}}. \tag{31}$$

If $\widetilde{U}$ is admissible at level $k+1$, then $L\widetilde{U}$ is admissible at level k. Its box diameter is $L\,\mathrm{diam}(\widetilde{U})$, and $N(L\widetilde{U}) \subset LN(\widetilde{U})$. The scaling rules on the common generators are $\psi(x) \mapsto L^{-1}\psi(x/L)$ and $\chi(x,x') \mapsto L^{-1-\alpha}\chi(x/L, x'/L)$. They preserve the chamber label and the period signs. Thus

$$S_L\iota_{L\widetilde{U}} = \iota_{\widetilde{U}}S_L. \tag{32}$$

Finally, the integrations agree on any local element. For W, expand in monomials. A monomial moment is zero unless its modes are paired; otherwise it is its ordering sign times $\nu^{-\#\mathrm{pairs}}$. The mode indices and the weight ν agree. At zero this weight is $1/48$, so the factor is $48^{\#\mathrm{pairs}}$; other modes integrate to 1. This proves

$$\iota_{\widetilde{Y}} \int \mathrm{d}\mu_I(W)\, F = \int \mathrm{d}\mu_I(W)\, \iota_{\widetilde{Y}} F. \tag{33}$$

The restricted photon law is a product of the same normalized Gaussian on $[-p_{0,k}, p_{0,k}]$ in each coordinate. Consequently its marginal on common coordinates is identical. With $d_V = 2 - 2L^{-3}$, its vacuum contribution on either torus is

$$\delta\varepsilon_k^0 = -d_V \log\left((2\pi)^{-1/2}\int_{-p_{0,k}}^{p_{0,k}} \exp(-z^2/2)\mathrm{d}z\right). \tag{34}$$

This number depends on e_k and L, and is independent of the side.

5.4. **Extraction and determinant comparison.** Suppose $\mathscr{E}_i \in K_k^x(T_i)$ agree in the sense of (26). For a small covering polymer $\widetilde{X}$, their values at $(0,0)$ have the same constant and quadratic coefficients. Their local extractions (67) are therefore the same: their integrated kernels and their common volume are the same. Fixing a covering cube identifies all small polymers containing its image on either torus. The sums (68) consequently give

$$m(\mathscr{E}_1) = m(\mathscr{E}_2), \qquad \varepsilon_{\mathrm{v}}(\mathscr{E}_1) = \varepsilon_{\mathrm{v}}(\mathscr{E}_2). \tag{35}$$

Subtracting the common constant and mass forms preserves local agreement of $R\mathscr{E}_i$. On a small polymer, its quadratic part also agrees; (31) proves agreement of $(R\mathscr{E}_i)^{\mathrm{nat}}$. On other polymers neither of these operations changes the relevant element.

The determinant terms agree on every flat polymer, not just on small ones. Let $X_i = \mathrm{pr}_i\,\widetilde{X}$ and suppose B_i have equal lifts on bonds meeting $\widetilde{X}$. For $k \geq 1$, set

$$O_1 = B_{k,r}\gamma_5 B_{k,r}, \qquad O_2 = Q_k J^* S_{k,r}(X_i, B_i) J Q_k^T(-B_i).$$

The trace in the determinant integral is

$$\mathrm{Tr}(O_1O_2) = \sum_{z_1,z_2 \in X_i \cap \mathbb{Z}^3} \mathrm{tr}\,(O_1(z_2,z_1)O_2(z_1,z_2)). \tag{36}$$

Here and below Tr is the trace on the whole finite function space, while tr is the spin trace. Indeed

$$O_2(z_1,z_2) = \int_{\Delta_{z_1}} \mathrm{d}x \int_{\Delta_{z_2}} \mathrm{d}x'\, \varphi_{B_i}(z_1,c) S_{k,r}(X_i,B_i)(x,x')\varphi_{B_i}(z_2,c')^{-1}.$$

Local polymer support confines x, x' to X_i, and Δ_z lies in the M-cube of z. Thus both unit sites in this trace lie in X_i. The rows of $B_{k,r}$ stay in their own L-block, so the intermediate sum in $O_1(z_2,z_1)$ also stays in X_i. All sums may now be formed on the lift $\widetilde{X}$; the two antiperiodic kernel signs cancel in each trace summand. The finite-row comparison and (29) make every factor the same on both tori.

At zero the required traces are

$$\mathrm{Tr}(F_0(B_i)S_{0,r}(X_i,B_i)) = \int_{X_i} \mathrm{d}x \int_{X_i} \mathrm{d}x' \ \mathrm{tr}\,(F_0(B_i)(x,x')S_{0,r}(X_i,B_i)(x',x)), \tag{37}$$

$$\mathrm{Tr}(\gamma_5 S_{0,r}(X_i,B_i)) = \int_{X_i} \mathrm{d}x \ \mathrm{tr}\,(\gamma_5 S_{0,r}(X_i,B_i)(x,x)). \tag{38}$$

Support again restricts all arguments to X_i. Both are the same covering sums with measure $1/48$ and all 48 chamber labels. The local F_0 and $S_{0,r}$ kernels agree, so the traces agree for every $r \geq 0$.

These pointwise identities can be integrated in r. The required domination follows from the primitive estimates. Since $P_B^2 = P_B$,

$$B_{k,r}\gamma_5 B_{k,r} = \gamma_5\big((b_k + \mathrm{i}\gamma_5 r)^{-2}(I - P_B) + (b_k + b_L + \mathrm{i}\gamma_5 r)^{-2}P_B\big).$$

In the component matrix norm, $|(a+\mathrm{i}\gamma_5 r)^{-1}|_c \leq \sqrt{2}(a^2+r^2)^{-1/2}$ for $a > 0$. Hence the first trace is bounded by a finite B_i, X_i dependent constant times $(b_k^2 + r^2)^{-1}$. The other factor is uniformly bounded by the local resolvent estimate and the two finite block transports. This majorant is integrable because $b_k > 0$. At zero, $|S_{0,r}(X_i,B_i)(x,x')|_c \leq C_w(1+r)^{-1}$. The first trace in (37) is integrated against $\mathrm{d}r/r$ on $[1,\infty)$, and (38) against $\mathrm{d}r$ on $[0,1]$; both are integrable. The kernels are Borel in r, so these are ordinary absolutely convergent scalar integrals.

The two unsymmetrized determinant primitives agree. Applying the argument at $-B_i, \bar{B}_i, -\bar{B}_i$ identifies all four terms in its symmetrization. It gives

$$E_k^{d,(1)}(X_1,B_1) = E_k^{d,(2)}(X_2,B_2). \tag{39}$$

Subtracting the values at $\mathscr{C}_i$ or at zero proves the corresponding equality for $\delta E_k^{\mathrm{det}}$ and E_k^{det}.

5.5. **Comparison of the local fluctuation sums.** Let m be real with $|m| \leq w_k^{\mathrm{m}}$, let $\|\mathscr{E}_i\|_k^x \leq w_k^{\mathrm{E}}$, and assume $\mathscr{E}_i = 0$ at zero and local agreement as above. Fix $A_1 \in \widetilde{R}_k^-(T_1)$ and $\mathscr{C}_1 \in \mathfrak{X}_k(T_1)$, and put $(A_2,\mathscr{C}_2) = p^*(A_1,\mathscr{C}_1)$. All small-charge and primitive hypotheses of the nonlinear step are imposed on both tori. In particular the contour radii satisfy $1 < r_2 \leq r_1$, $r_4 > 1$, and the domain inclusions of Proposition 5.1 of [NL] hold. We compare the terms of that step on a connected nonempty $\widetilde{U} \subset \widetilde{Y}$.

Assign common coordinate values $z_\Upsilon \in [-p_{0,k}, p_{0,k}]$ to the photon basis elements contained in $\widetilde{U}$, and set all other coordinate values to zero on each torus. Assign common parameter values of modulus at most M^{α_0} to the parameter cubes contained in $\widetilde{U}$, and set all other parameters to zero. These data will be denoted by ζ_i, s_i. They belong to the respective box and polydisc because indices on the lift are injective.

The fine fluctuation fields have the same lifts on $\widetilde{U}^{\sim(1+d_U)}$:

$$Z_1(s_1')(b) = Z_2(s_2')(b). \tag{40}$$

To check the successive supports in their composition, write it in kernels,

$$Z_i(s_i')(b) = \sum_{b',\Upsilon,\Upsilon'} H_k^{\mathrm{ph},(i)}(s_i')(b,b')C_i(b',\Upsilon)C_k^{1/2,(i)}(s_i')(\Upsilon,\Upsilon')\zeta_i(\Upsilon').$$

At such a row b, proposition 5.2 with $j = 1 + d_U$ confines b' to $\widetilde{U}^{\sim 2d_U}$ and identifies the row kernels. The coordinate row has columns in the M-cubes of the two ends of b', hence in $\widetilde{U}^{\sim(1+2d_U)}$; those finite rows agree. The square-root row, with $j = 1 + 2d_U$, confines Υ' to $\widetilde{U}^{\sim 3d_U}$. All these sets lie in $N(\widetilde{U})$, and the last coordinate values agree there. The two compositions are therefore the same finite covering sum. At zero the first row is diagonal and the sum over b' has one term.

Put $B_i = A_i + \mathscr{C}_i$ and $B_i(t) = B_i + tZ_i(s_i')$. For $|t| \leq r_1$ these fields lie in $\widetilde{\mathcal{G}}_k(T_i)$, and $A_i + tZ_i(s_i') \in \widetilde{R}_k(T_i)$. They have equal lifts on the bonds just compared. The fermion mean-field rows at sites at distance at most η_k from $\widetilde{U}$ consequently agree, as do the Γ_k rows at unit sites in $\widetilde{U}$

and the $M_k(X_i, B_i(t))$ kernels for lifted subpolymers $\widetilde{X} \subset \widetilde{U}$. Here the parameters vanish outside U_i. The row-vanishing and component argument in the proof of Proposition 5.3 of [NL] confines their columns to U_i itself: a nonzero mean-field row has its active parameter component attached to that set; the Γ_k row starting in this component has the same confined columns. For $k \geq 1$, the parameter-independent term $\mathcal{B}_{k,0}$ in Γ_k also preserves U_i, a union of partition M-cubes. Thus all these fields are polynomials in the common generators over $\widetilde{U}$.

For example, at a relevant chamber site,

$$(H_k^{(i)}(s_i'', B_i(t))\Psi)_\beta(x) = \sum_{y \in \widetilde{U} \cap \mathbb{Z}^3, \beta'} H_k^{(i)}(s_i'', B_i(t))(x,y)_{\beta\beta'} \Psi(y, \beta', 0)$$

for $k \geq 1$, with the analogous barred sum. Their coefficients agree. At every pair admissible for $\widetilde{X}$, applying the common finite difference row preserves this equality. The differences between these fields at $t = 1$ and $t = 0$ are precisely the ordinary and pair components of $J_k^{\#}\Psi$ used in G_2, G_3. Both values are in the disk because $r_1 > 1$. Moreover locality of $R\mathscr{E}_i$ gives equality of its values at the pairs $(A_i + tZ_i, \mathscr{C}_i)$ on each subpolymer. These pairs need only agree on its read bonds, by the criterion following (26).

It follows that the three first-localization contour integrands agree under $\iota_{\widetilde{U}}$. For G_1, this is the common local M_k bilinear. For G_2, addition of the common odd field $tJ_k^{\#}\Psi$ to each generator commutes with an algebra homomorphism; apply it to the equal local activity. For G_3, the same substitution applies to the equal mass form, and $m' = m + m(\mathscr{E}_i)$ is common by (35). Their contour radii and scalar contour measures are the same. Integration thus gives

$$\iota_{\widetilde{U}} G_a^{(1)}(s_1, X_1) = G_a^{(2)}(s_2, X_2), \qquad a = 1, 2, 3. \tag{41}$$

At zero $G_2 = G_3 = 0$ and no auxiliary field occurs. G_1 is the first-step vertex; its two factors $\exp(\pm iqZ_i(c, c_3)) - 1$ agree by (40), with its common coefficient $1/96$ and the common outward pair. This proves (41) at that level too.

The finite localization commutes with this equality. Indeed the pairs (X, P) with $Y'(X, P) = U_i$ correspond bijectively to lifted pairs $(\widetilde{X}, \widetilde{P})$ inside $\widetilde{U}$ having $Y'(\widetilde{X}, \widetilde{P}) = \widetilde{U}$. If a torus pair has that union, its polymer and all its parameter cubes are contained in U_i, so the unique lifts give the inverse correspondence. The identities for P_0, Y' proved above show that it preserves the parameter prescriptions. Its summand is therefore

$$\int_{[0,1]^{\widetilde{P}}} \mathrm{d}u \prod_{\mathrm{pc} \in \widetilde{P}} \partial_{u_{\mathrm{pc}}} G_a^{(i)}(s_i[u], \mathrm{pr}_i \widetilde{X}), \qquad a = 1, 2, 3, \tag{42}$$

where $s_i[u]$ is 1 on $P_0(\widetilde{X})$, u on $\widetilde{P}$, and zero elsewhere. Identity (41) holds throughout the parameter polydisc. Since $M^{\alpha_0} > 1$, the integration cube is in its interior, and coefficient differentiation and integration preserve the equality. This proves equality of $\mathrm{Loc}\, G_a(U_i)$.

The reconstruction substitution replaces each $\Psi(y)$ by the same two-piece row $T^{(1)}\psi + T^{(2)}\chi$. Both rows, including their boundary membership indicators, have already been compared. Their resulting ordinary generators are over $\widetilde{U}$, and their boundary pairs are admissible for it. Equality on generators proves equality of the substitution homomorphisms. We obtain

$$\iota_{\widetilde{U}} E_{\leq 3}^{(1)}(U_1) = E_{\leq 3}^{(2)}(U_2). \tag{43}$$

At zero no reconstruction substitution is made.

For the second localization, the common fermion field is

$$\mathrm{Wc} = H_k(s'', B)\Gamma_k(s'', B)W, \qquad \overline{\mathrm{Wc}} = \bar{H}_k(s'', B)\bar{W}.$$

For $k \geq 1$ both are sums over the common unit sites of $\widetilde{U}$. For $k = 0$ they are

$$\mathrm{Wc}(x) = d(c)\frac{1}{48} \sum_{x' \text{ over } \widetilde{U}} \Gamma_0(s'', B)(x, x')W(x'), \qquad \overline{\mathrm{Wc}}(x) = d(c)\bar{W}(x),$$

where the common diagonal factor $d(c)$ vanishes outside U_i. These formulas preserve all chamber labels and the weight $1/48$. The kernel equalities proved above, followed by the common pair difference rows, identify $W_k^{\#}$ at every index used by a subpolymer.

The first-localized activity on a subpolymer reads only photon coordinates supported there. Hence (43), proved with coordinates supported on that subpolymer, still holds when coordinates are supported on the larger $\widetilde{U}$. Adding the common $W_k^{\#}$ to its generators proves equality of $G^W_{\leq 3}$. The same addition to $R\mathscr{E}_i$ and to the mass form proves equality of the contour integrands of $G_4, \bar{G}_5$. The second localization uses only π_k; the correspondence and derivative argument (42) apply with this one kind. Consequently

$$\iota_{\widetilde{U}} F_1(U_1) = F_2(U_2). \tag{44}$$

We now pass to the local connected logarithm. Its finite Mayer activity is

$$K_i(X_i) = \sum_{\{\widetilde{X}_a\}: \bigcup_a \widetilde{X}_a = \widetilde{X}} \prod_a \big(\exp(F_i(\mathrm{pr}_i \widetilde{X}_a)) - 1\big),$$

where the $\widetilde{X}_a$ are distinct connected subpolymers of $\widetilde{X}$. The products are even. Their index sets and factors are the same on both tori by (44); coordinate locality allows their evaluation with common coordinates on $\widetilde{X}$. For the integrated activity $H_i(X_i)$, (33) identifies the Wick integrals. The other photon coordinates integrate to 1, leaving the same product of restricted normal laws over the coordinates supported in $\widetilde{X}$. Hence $\iota_{\widetilde{X}} H_1(X_1) = H_2(X_2)$.

The logarithm has the expansion

$$E_k^{\#,(i)}(X_i) = \sum_{n \geq 1} \frac{1}{n!} \sum_{\bigcup_{a=1}^n \widetilde{X}_a = \widetilde{X}} u^T(\mathrm{pr}_i \widetilde{X}_1, \dots, \mathrm{pr}_i \widetilde{X}_n) \prod_{a=1}^n H_i(\mathrm{pr}_i \widetilde{X}_a).$$

Projection preserves incompatibility inside this lift, so the graph determining u^T is the same. Every summand agrees under the common algebra map. Both series converge absolutely in the weighted norm by Proposition 6.1 of [NL]. The map $\iota_{\widetilde{X}}$ preserves that norm, so equality passes to their sums:

$$\iota_{\widetilde{X}} E_k^{\#,(1)}(X_1; A_1, \mathscr{C}_1) = E_k^{\#,(2)}(X_2; A_2, \mathscr{C}_2). \tag{45}$$

5.6. **Agreement of the step after reblocking.**

Proof of Theorem B. The scalar conclusions are (35) and (34). For the outgoing activity it suffices to compare a level $k+1$ connection pair with its pullback. Let $\widetilde{U}$ be connected and admissible there, put $Y_i = \mathrm{pr}_i' \widetilde{U}$, and set $\widetilde{Y} = L\widetilde{U}$. The latter is a connected union of LM-cubes, hence also of M-cubes, admissible at level k. Its neighborhood is contained in $LN(\widetilde{U})$.

If $(\mathscr{A}', \mathscr{C}')$ satisfies the outgoing-domain hypotheses at level $k+1$, its scaled pair lies in $\widetilde{R}_k^- \times \mathfrak{X}_k$ by (84). Scaling and pullback commute on covering bonds: both give $L^{-1/2}B'(b/L)$. Denote the scaled pairs by $(A_i, \mathscr{C}_i)$. They satisfy exactly the hypotheses used in (45).

The polymers with $\bar{X}^L = LY_i$ are precisely $\mathrm{pr}_i \widetilde{X}$, where $\widetilde{X}$ is connected in $\widetilde{Y}$ and its LM-closure is $\widetilde{Y}$. Indeed $X \subset \bar{X}^L$ gives its unique lift inside $\widetilde{Y}$. Closure commutes with projection, and lies in $\widetilde{Y}^{\sim(L-1)}$. Projection is injective on that enlargement, so equality of its projected closure with $\mathrm{pr}_i \widetilde{Y}$ forces equality of the lifted closure with $\widetilde{Y}$. This also proves the converse and shows that the reblocking index set is common.

In that index set the outgoing activity is

$$\mathscr{E}_i^{+}(Y_i;\mathscr{A}_i',\mathscr{C}_i') = \sum_{\substack{\widetilde{X}\subset\widetilde{Y}\ \text{connected}\\ \overline{\widetilde{X}}^{\,L}=\widetilde{Y}}} S_L\Big[(R\mathscr{E}_i)^{\mathrm{nat}}(\mathrm{pr}_i\,\widetilde{X};A_i,\mathscr{C}_i) + \delta E_k^{\mathrm{det},(i)}(\mathrm{pr}_i\,\widetilde{X};A_i,\mathscr{C}_i) + E_k^{\#,(i)}(\mathrm{pr}_i\,\widetilde{X};A_i,\mathscr{C}_i)\Big]. \tag{46}$$

Each of the three entries agrees under $\iota_{\widetilde{Y}}$, by extraction and adjustment, (39), and (45), respectively. The entries belong to its ordinary and pair subalgebra. Equation (32) identifies their scaled values, so the sums agree under $\iota_{\widetilde{U}}$. Locality of the outgoing family, already proved in the step theorem, then extends the comparison from pullback pairs to pairs with equal local lifts.

The external scalar recursion has the same reblocking index set. Its incoming scalar term agrees by hypothesis, and its local determinant term agrees by (39) at $(\mathscr{C}_1,\mathscr{C}_2)$ and at $(0,0)$. Its two sums therefore agree.

The argument without an external slot uses precisely these local operations with $B_i = A_i$. The determinant difference is then $E_k^{\det}(X_i,A_i) = E_k^{d}(X_i,A_i) - E_k^{d}(X_i,0)$. Extraction, addition of odd fields, finite localization and integration do not require that slot. Their same domain and convergence assertions were established for this case in the nonlinear-step construction. Thus the three bracketed terms in (46) agree in that case too, and scaling gives the conclusion. Restriction of a family with a slot to $\mathscr{C}=0$ also commutes with every operation in the displayed formula.

At $k=0$ the incoming activity and its extracted terms are zero. The comparison is for the local determinant difference and the first correction, with chamber generators, traces and Wick moments of weight $1/48$, as explicitly included above. □

6. Physical scales and stopping levels

The logarithmic charge scale determines where the finite-volume iteration stops. We express the stopping rule in terms of depth below the physical mass scale. This removes the cutoff from the terminal physical quantities and gives the side margins needed for comparison at every preceding step. The stopping-scale calculation follows [QED, §16.1].

6.1. **Physical scales and stopping levels.** Fix $\bar m \in (0,1]$ and put $h = L^{\lfloor \log_L(1/\bar m)\rfloor}$. For a cutoff $N \ge 0$, define

$$\eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \qquad e_j = e\sqrt{a_j}, \qquad \mu_j = \bar m a_j. \tag{47}$$

Thus a_j is the physical length of one unit at level j, and $a_K = h$. At zero the charge is written $q = e\sqrt{\eta}$; the fixed charge ceilings $\bar e_0, \bar e_1$ retain their earlier meanings. A formula with variable index j uses e_j for the running charge, including q when $j=0$. For $0 \le j \le K$ set

$$v_j = -\log e_j, \qquad r_j = \min\{L^i : i \in \mathbb{N}\cup\{0\},\ L^i \ge v_j^r\}, \qquad p_j = v_j^p, \quad p_{0,j} = v_j^{p_0}.$$

The integers r, p, p_0 and the constants of the preceding estimates are fixed. Here $r \ge 2$, L is odd with $L \ge 13$, and M is a power of L with $M \ge L^2$. We assume the hypotheses of the primitive and step estimates, and use a ceiling $\bar e_1 \le \min(\bar e_0, 1/3)$ such that, for all $v \ge -\log \bar e_1$,

$$\begin{gathered} v \ge 8, \qquad \Big(1 + \frac{3\log L}{2v}\Big)^r < L, \qquad v^r > 31L + 16, \\ (c_{\mathrm{fl}} + 2)\exp(-\varepsilon v) \le \frac{1}{16}(1 - L^{-\varepsilon/2}). \end{gathered} \tag{48}$$

In addition $0 < \varepsilon \le 1/200$, $c_{\mathrm{fl}} \ge 1$, $\bar e_0^{\varepsilon} \le 1/4$, $c_{\mathrm{fl}} L^{-3/8+6/200} \le 1/16$, and $L^{-5/8} \le 1/4$. The construction below applies whenever $0 < e\sqrt{h} \le \bar e_1$.

The physical top side and reference side are

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L = hL^{T^*(e)}, \qquad T^*(e) = \log_L(Mr_K) - 1. \tag{49}$$

They are independent of N, since $v_K = -\log(e\sqrt{h})$. The admissible physical sides for the flow are

$$\mathcal{T}(e) = \{hL^T : T \in \mathbb{N} \cup \{0\},\ hL^T \le R_{\mathrm{top}}\}.$$

For $R \in \mathcal{T}(e)$ put $\ell_j(R) = R/a_j$, $T_j(R) = \eta_j\mathbb{Z}^3/\ell_j(R)\mathbb{Z}^3$ and $\mathfrak{n}_j = \log_L(\ell_j/(Mr_j))$. The allowed unit sides are $\mathrm{Cl} = \{\ell : \ell/(LM)$ is an odd integer at least $13\}$. Their occurrence at every required scale is part of the following arithmetic argument.

Proposition 6.1. For each $R \in \mathcal{T}(e)$ there is a finite integer $N_{\min}(R,e) \ge 0$ such that, when $N \ge N_{\min}(R,e)$, the first level with $\mathfrak{n}_j = 0$ exists and satisfies $1 \le k_*(R) \le K$. Its physical quantities $a_*, e_*, \mu_*, \ell_*, r_*$ are independent of N in this range. For $j \le k_*(R)$,

$$\begin{gathered} \ell_j \in \mathrm{Cl}, \qquad \ell_j \ge Mr_j, \qquad r_j \ge r_* > 31L + 16, \\ v_j \ge v_* \ge v_K, \qquad 0 < e_j \le e_* \le e_K \le \bar{e}_1, \qquad 0 < \mu_j \le \mu_* \le \bar{m}h \le 1. \end{gathered} \tag{50}$$

For $j < k_*(R)$ one has $\ell_j \ge LMr_j$, $\ell_{j+1} = \ell_j/L \in \mathrm{Cl}$, $e_{j+1} = \sqrt{L}e_j \le \bar{e}_0$ and $0 < \mu_j = \mu_{j+1}/L \le 1/L$. At the stop,

$$\ell_* = Mr_*, \qquad Mv_*^r \le \ell_* < LMv_*^r, \qquad \mu_*^{-1} < L\ell_*, \qquad e_*^2\ell_* = e^2R. \tag{51}$$

If $R \le R'$ in $\mathcal{T}(e)$, then $N_{\min}(R',e) \le N_{\min}(R,e)$ and $k_*(R) \le k_*(R')$ for a common cutoff above both stated thresholds. ∎

Proof. Use the depth $s = K - j$. Its quantities are

$$\widehat{v}_s = v_K + \tfrac{s}{2}\log L, \qquad \widehat{r}_s = \min\{L^i : i \ge 0,\ L^i \ge \widehat{v}_s^r\}, \qquad \widehat{n}_s(R) = T + s - \log_L(M\widehat{r}_s).$$

They do not depend on N. At depth $s = K - j$ they equal $v_j, r_j, \mathfrak{n}_j$, while $a_j = hL^{-s}$ and $\ell_j = L^{T+s}$. Since $\widehat{v}_s \ge v_K$, the scalar condition gives

$$\widehat{v}_{s+1}^r = \widehat{v}_s^r\left(1 + \frac{\log L}{2\widehat{v}_s}\right)^r \le L\widehat{v}_s^r \le L\widehat{r}_s.$$

Thus $\widehat{r}_{s+1} \in \{\widehat{r}_s, L\widehat{r}_s\}$ and $\widehat{n}_{s+1} - \widehat{n}_s \in \{0,1\}$. Also $\widehat{n}_0 \le 0$ because $R \le R_{\mathrm{top}}$. Minimality of the rounded power gives $\widehat{r}_s < L\widehat{v}_s^r$ and hence

$$\widehat{n}_s > T + s - 1 - \log_L M - r\log_L(v_K + \tfrac{s}{2}\log L) \longrightarrow \infty.$$

The integer sequence therefore has a largest zero. Denote its index by $s_*(R,e)$. It is nonpositive up to that index and at least 1 afterwards.

The condition $\ell_0 \ge LMr_0$ is $\widehat{n}_K \ge 1$, equivalently $K > s_*$. Its exact threshold is

$$N_{\min}(R,e) = \max\{0, s_*(R,e) + 1 - \log_L h\}, \qquad k_*(R) = K - s_*(R,e). \tag{52}$$

This gives $1 \le k_* \le K$ and $a_* = hL^{-s_*}$, proving cutoff independence of the terminal quantities. Increasing T increases every $\widehat{n}_s$ by the same integer; thus it decreases s_*, decreases $N_{\min}$ and increases k_*.

For $j \le k_*$, the depth satisfies $K - j \ge s_*$. Monotonicity gives the asserted ranges of $\mathfrak{n}_j, r_j, v_j, e_j$, and the running mass increases with j. Moreover $\bar{m}h = L^{\lfloor \log_L(1/\bar{m})\rfloor - \log_L(1/\bar{m})} \in (1/L, 1]$. Since $\mu_*\ell_* = \bar{m}R \ge \bar{m}h > 1/L$, we obtain $\mu_*^{-1} < L\ell_*$. The rounded-power inequality at v_* gives the two bounds for ℓ_* in (51); its last identity uses $e_*^2 = e^2a_*$.

Finally $r_* \ge v_*^r > 31L + 16 > L$ implies $r_* \ge L^2$, as it is a power of L. At every required level, $\ell_j/(LM) = r_jL^{\mathfrak{n}_j - 1}$ is a positive power of the odd integer $L \ge 13$. This proves $\ell_j \in \mathrm{Cl}$. Before the stop $\mathfrak{n}_j \ge 1$, giving the stronger side bound. The mass and charge bounds for the step follow at $j + 1$. □

On R^* define the reference mesh threshold

$$N_{\mathrm{ref}}(e) = N_{\min}(R^*(e), e). \tag{53}$$

The stop on this torus has depth $s_* \in \{1, 2\}$. Indeed, for $s = 1, 2, 3$ the stronger inequality in (48) gives $\widehat{r}_s \in \{r_K, Lr_K\}$. On R^*, $\widehat{n}_s = s - 1 - \log_L(\widehat{r}_s/r_K)$. Its values at $s = 0, 1, 2, 3$ are respectively -1, an element of $\{-1, 0\}$, an element of $\{0, 1\}$, and an element of $\{1, 2\}$. The zero-step-or-one-step rule forces the last zero to be 1 or 2. On R_{top} the sequence is larger by one, so its last zero is 0 or 1. Consequently

$$\begin{gathered} k_*(R^*) \in \{K-1, K-2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad r_* \in \{r_K, Lr_K\}, \\ k_{\mathrm{top}} \in \{K, K-1\}, \qquad L^{-3} < \mu_* \le L^{-1} \quad \text{on } R^*. \end{gathered} \tag{54}$$

All these physical terminal values are independent of $N \ge N_{\mathrm{ref}}(e)$.

The scale proposition verifies every side, mass and charge hypothesis of the step theorem on R_{top} and on the smaller tori it covers with odd degree. At each level use the top system of section 4.3; on a smaller torus use its projection from proposition 4.2. These are the actual mean-field, covariance, local resolvent and photon families constructed earlier, with the same common estimates. The ratio R_{top}/R is a power of the odd integer L, and $k_*(R) \le k_{\mathrm{top}}$. Thus the required top families exist whenever a smaller-volume step is used. Furthermore $\ell_{j+1}(R) \ge Mr_{j+1} > (31L + 16)M$; the next-side hypothesis of the cover comparison holds at every such step.

7. The terminal conditions and the common pair

We give the continuation and tuning arguments of [QED, §§16.2–16.4].

7.1. **Continuation within the mass range.**

Write the mass and activity bounds as

$$w_j^{\mathrm{m}} = e_j^{3/4-8\varepsilon}, \qquad w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}.$$

For a real pair $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$, the initial vacuum is $\varepsilon_{\mathrm{in}}^{\mathrm{v}}$, the initial member of the mass sequence is m_{in}, and $E_0 = \mathscr{E}_0 = \mathfrak{P}_0 = 0$. The variable-index notation m_j includes this initial member. The recursion is

$$m_{j+1} = L(m_j + m(E_j)), \qquad \varepsilon_{j+1}^{\mathrm{v}} = L^3(\varepsilon_j^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(E_j) + \delta\varepsilon_j^0), \tag{55}$$

$$E_{j+1} = \mathcal{L}(RE_j)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m_j, E_j), \tag{56}$$

$$\mathscr{E}_{j+1} = \mathcal{L}(R\mathscr{E}_j)^{\mathrm{nat}} + \mathcal{L}\delta E_j^{\mathrm{det}} + E_j^*(m_j, \mathscr{E}_j). \tag{57}$$

The scalar family $\mathfrak{P}_j$ follows (14). In the last line the correction has the external slot; in (56) it has none. These are the corresponding maps proved earlier. The trajectory exists up to a level if every preceding $(m_j, \mathscr{E}_j)$ lies in $\mathrm{Ball}_j^x(4)$. It is controlled when it exists to $k_*(R)$ and $|m_j| \le w_j^{\mathrm{m}}$ throughout, including the stop.

The normalized estimates used below are

$$\begin{gathered} L^{-1}\frac{w_{j+1}^{\mathrm{m}}}{w_j^{\mathrm{m}}} = L^{-5/8-4\varepsilon} \le \tfrac{1}{4}, \qquad c_{\mathfrak{fl}} e_j^{1/2}\frac{w_j^{\mathrm{E}}}{w_j^{\mathrm{m}}} = c_{\mathfrak{fl}} e_j^{\varepsilon}, \\ c_{\mathfrak{fl}} L^{-1/4+2\varepsilon}\frac{w_j^{\mathrm{E}}}{w_{j+1}^{\mathrm{E}}} = c_{\mathfrak{fl}} L^{-3/8+11\varepsilon/2} \le \tfrac{1}{16}, \\ \frac{e_{j+1}^{1/4-2\varepsilon}}{w_{j+1}^{\mathrm{E}}} \le e_{j+1}^{\varepsilon}, \qquad \frac{e_j^{1/4-6\varepsilon}}{w_{j+1}^{\mathrm{E}}} = L^{-1/8+3\varepsilon} e_{j+1}^{\varepsilon} \le e_{j+1}^{\varepsilon}. \end{gathered} \tag{58}$$

Their first and third bounds use the fixed inequalities stated above; their last two use $e_{j+1} < 1$ and $\varepsilon \le 1/200$. The charge condition also gives

$$\sum_{j=0}^{k_*} (c_{\mathrm{fl}} + 2) e_j^{\varepsilon} \le (c_{\mathrm{fl}} + 2) e_*^{\varepsilon} (1 - L^{-\varepsilon/2})^{-1} \le \tfrac{1}{16}. \tag{59}$$

In particular $c_{\mathrm{fl}} e_j^{\varepsilon} \le 1/16$, $2e_j^{\varepsilon} \le 1/16$, and $e_j \le e_j^{\varepsilon} \le 1/48$.

Proposition 7.1. If the recursively produced masses satisfy $|m_i| \le w_i^{\mathrm{m}}$ for $i < j \le k_*(R)$, then the trajectory is defined to j and

$$\|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le \frac{32}{15} e_j^{\varepsilon} w_j^{\mathrm{E}} \le 4 e_j^{1/4-6\varepsilon} \le w_j^{\mathrm{E}}. \tag{60}$$

The data satisfy $E_j = \mathscr{E}_j|_{\mathscr{C}=0}$, $\mathfrak{P}_j(X, 0) = 0$, and belong to their real mass, vacuum and activity spaces. ■

Proof. At zero all activities and scalar families vanish. Assume the assertion through $i < j$. The mass condition and $4e_i^{\varepsilon} \le 1$ put the input in the unit ball, so Theorem A of [NL] applies at i. The scaled connection pair lies in $\widetilde{R}_i^- \times \mathfrak{X}_i$, so every local determinant and correction in the recursion is defined. The scalar external sum is finite and uses $\mathscr{C}_L \in \mathfrak{X}_i$. The step theorem preserves the real spaces and commutes with restriction to $\mathscr{C} = 0$. Consequently the coupling maps computed from E_i or $\mathscr{E}_i$ coincide. The scalar recursion vanishes at zero because each $E_i^{\mathrm{det}}(X, 0)$ vanishes.

The activity bound of the current step, divided by w_{i+1}^{E} and combined with (58), gives

$$\frac{\|\mathscr{E}_{i+1}\|_{i+1}^x}{w_{i+1}^{\mathrm{E}}} \le \frac{1}{16} \frac{\|\mathscr{E}_i\|_i^x}{w_i^{\mathrm{E}}} + 2e_{i+1}^{\varepsilon}.$$

Starting from zero, the bound $(32/15)e_i^{\varepsilon}$ is preserved since $e_i \le e_{i+1}$ and $(32/15)/16 + 2 = 32/15$. Restriction does not increase the activity norm. This proves (60) and the induction.

At the first step the formulas use $192\ell_0^3$ incoming pairs with weight $1/48$ and $4\ell_1^3$ outgoing pairs, as proved in the exact density identity. Its coupling maps are $m_1 = Lm_{\mathrm{in}}$ and $\varepsilon_1^{\mathrm{v}} = L^3(\varepsilon_{\mathrm{in}}^{\mathrm{v}} + \delta\varepsilon_0^0)$; its activity is the local determinant term plus the first correction. The normalized estimate has zero incoming-activity term and the same two source bounds. Thus the induction includes that step. □

7.2. **The terminal conditions and the fixed point.**

Theorem C. On any $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$, there is exactly one real controlled pair whose trajectory has $m_{k_*} = 0$ and $\varepsilon_{k_*}^{\mathrm{v}} = 0$. Along its trajectory,

$$|m_j| \le \tfrac{1}{8} w_j^{\mathrm{m}}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^{\mathrm{v}}| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}. \tag{61}$$

The vacuum at every level is

$$\varepsilon_j^{\mathrm{v}} = -\sum_{i=j}^{k_*-1} L^{-3(i-j)} (\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon_i^0). \tag{62}$$

■□

Proof. Take the finite product of the real mass coordinates and real activity Banach spaces, with norm

$$\|\mathbf{z}\|_{\mathrm{fl}} = \max_{0 \le j \le k_*} \max\left(\frac{|m_j|}{w_j^{\mathrm{m}}}, \frac{\|E_j\|_j}{w_j^{\mathrm{E}}} \right), \qquad \mathbf{z} = (m_j, E_j)_{0 \le j \le k_*}.$$

The closed subset $\mathcal{B}$ defined by these quotients being at most 1 and $E_0 = 0$ is nonempty and complete. Define a self-map by

$$\widehat{m}_j = L^{-1} m_{j+1} - m(E_j) \quad (j < k_*), \qquad\qquad \widehat{m}_{k_*} = 0,$$

$$\widehat{E}_0 = 0, \qquad \widehat{E}_{j+1} = \mathcal{L}(RE_j)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m_j, E_j). \tag{63}$$

All maps here are the actual maps without an external slot; their inputs belong to the unit ball. Extraction, adjustment followed by reblocking, the determinant activity and the nonlinear correction are defined on that ball and preserve the real space.

We verify that the map preserves $\mathcal{B}$. The extraction estimate implies $|m(E_j)| \le c_{\mathfrak{fl}} e_j^{1/2} \|E_j\|_j$. Equations (58) and (59) give

$$\frac{|\widehat{m}_j|}{w_j^{\mathrm{m}}} \le \tfrac{1}{4} + c_{\mathfrak{fl}} e_j^{\varepsilon} \le \tfrac{5}{16}.$$

For the activity, the contraction, determinant and correction estimates give

$$\frac{\|\widehat{E}_{j+1}\|_{j+1}}{w_{j+1}^{\mathrm{E}}} \le \tfrac{1}{16} + 2e_{j+1}^{\varepsilon} \le \tfrac{1}{8}.$$

The fixed end components are zero. Hence the map takes $\mathcal{B}$ into itself.

For two sequences, let $d_j^{\mathrm{m}}, d_j^{\mathrm{E}}$ be the corresponding normalized differences. Linearity of extraction gives

$$\widehat{d}_j^{\mathrm{m}} \le \tfrac{1}{4} d_{j+1}^{\mathrm{m}} + c_{\mathfrak{fl}} e_j^{\varepsilon} d_j^{\mathrm{E}} \le \tfrac{5}{16} \|\mathbf{z} - \widetilde{\mathbf{z}}\|_{\mathfrak{fl}}.$$

Adjustment followed by reblocking is linear in the activity at fixed connection. The local determinant term is independent of the sequence. The correction's coupling estimate (93) of [NL] therefore gives

$$\widehat{d}_{j+1}^{\mathrm{E}} \le \tfrac{1}{16} d_j^{\mathrm{E}} + e_{j+1}^{\varepsilon} (d_j^{\mathrm{m}} + d_j^{\mathrm{E}}) \le \tfrac{1}{8} \|\mathbf{z} - \widetilde{\mathbf{z}}\|_{\mathfrak{fl}}.$$

The whole map contracts by 5/16. Banach's fixed point theorem [Bri23, Theorem 8] gives a unique fixed point in $\mathcal{B}$. Its equations are precisely the mass and activity recursions with $m_{k_*} = 0$.

Let m_{in} be its initial mass coordinate and choose

$$\varepsilon_{\mathrm{in}}^{\mathrm{v}} = -\sum_{i=0}^{k_*-1} L^{-3i} (\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon_i^0).$$

This pair is real. Its full trajectory, including the external slot, agrees with the fixed-point mass and activity at every level: agreement at zero is the initial condition; at the next level proposition 7.1 makes the full step defined, and the fixed-point equations identify its mass and restricted activity. The induction reaches k_* and proves control. Solving the scalar vacuum recursion backwards gives

$$\varepsilon_j^{\mathrm{v}} = L^{-3(k_*-j)} \varepsilon_{k_*}^{\mathrm{v}} - \sum_{i=j}^{k_*-1} L^{-3(i-j)} (\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon_i^0).$$

At $j = 0$ its comparison with the chosen input forces $\varepsilon_{k_*}^{\mathrm{v}} = 0$, giving (62). Conversely any controlled pair with the two terminal zeros has its mass/activity sequence in $\mathcal{B}$ by the continuation proposition. It is the same fixed point. Its initial vacuum is then the same scalar sum. This proves uniqueness of the pair.

For the sharper mass bound put $B_j = |m_j|/w_j^{\mathrm{m}}$. The backward mass equation and the unit activity bound imply $B_j \le B_{j+1}/4 + 1/16$, while $B_{k_*} = 0$. Summing gives $B_j \le (1/16) \sum_{s \ge 0} 4^{-s} = 1/12 \le 1/8$. The activity bound in (61) is the continuation estimate.

It remains to bound the vacuum sum. Extraction gives $|\varepsilon_{\mathrm{v}}(E_i)| \le 4 c_{\mathfrak{fl}} e_i^{1/4-6\varepsilon}$. For $i \ge j$ the associated summand obeys

$$L^{-3(i-j)} e_i^{1/4-6\varepsilon} = e_j^{1/4-6\varepsilon} L^{-(i-j)(23/8+3\varepsilon)} \le e_j^{1/4-6\varepsilon} 16^{-(i-j)}.$$

Thus its sum is at most $(64/15)c_{\mathrm{fl}}e_j^{1/4-6\varepsilon}$. The box normalization satisfies $0 \le \delta\varepsilon_i^0 \le e_i^7$ by the step estimate. Its remaining sum is bounded by

$$\sum_{i=j}^{k_*-1} L^{-3(i-j)} e_i^7 = e_j^6 \sum_{i=j}^{k_*-1} e_i \le e_j^6 e_* (\sqrt{L}-1)^{-1} \le \tfrac{1}{48} e_j^{1/4-6\varepsilon}.$$

Here $\sqrt{L} \ge 2$ and $e_* \le 1/48$. Since $c_{\mathrm{fl}} \ge 1$, $64/15 + 1/48 = 343/80 < 5$, giving the claimed bound. The sums are empty at the stop.

At zero $E_0 = 0$. The backward mass equation reads $m_{\mathrm{in}} = L^{-1}m_1$, and the forward activity is $\mathcal{L}E_0^{\mathrm{det}} + E_0^*(m_{\mathrm{in}}, 0)$. The determinant and correction estimates used above include their full chamber trace and the first weighted fluctuation integral. The initial vacuum sum contains $\delta\varepsilon_0^0$. No additional coupling is selected by the fixed point. □

7.3. **One pair on all tori of the construction.** For $N \ge N_{\min}(R_{\mathrm{top}}, e)$ define

$$P_N(e) = (m_{0,N}(e), \varepsilon^{\mathrm{v}}_{0,N}(e))$$

to be the pair of theorem C on the prescribed side $R_{\mathrm{top}}(e)$. The dependence on the fixed $\bar{m}$ and fixed construction parameters is suppressed. For lower cutoffs set $m_{0,N}(e) = 0$.

Theorem D. The pair $P_N(e)$ is controlled on every $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$. Up to its stop, its mass and vacuum sequence agree with those on the top torus; its activity and external scalar families agree there on flat polymers. For every such level,

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^{\mathrm{v}}| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}. \tag{64}$$

The real mass sequence is independent of the side datum and of the twist. At all $N \ge 0$,

$$|m_{0,N}(e)| \le \tfrac{1}{8} q^{3/4-8\varepsilon} < \tfrac{1}{8}, \qquad q = eL^{-N/2}. \tag{65}$$

The total lattice mass is $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$. ■□

Proof. The pair is defined because $N_{\min}(R_{\mathrm{top}}, e) \le N_{\min}(R, e)$. The two trajectories have the same initial couplings and zero initial activities. Suppose they agree through $j < k_*(R)$. The top mass is bounded by $w_j^{\mathrm{m}}/8$; the same bound on the smaller side permits continuation to $j+1$ by proposition 7.1. Both inputs are now in the unit balls, are real, and have locally agreeing activity and scalar families. The scale proposition verifies their primitive, domain and next-side comparison hypotheses. theorem B gives the same extracted mass and vacuum, including the same $\delta\varepsilon_j^0$, and local agreement of the outgoing activity and scalar family. The scalar recursions therefore give the same $m_{j+1}, \varepsilon_{j+1}^{\mathrm{v}}$. Restriction to zero external slot preserves agreement of E_{j+1}. This proves the induction up to $k_*(R)$.

The common mass bound follows from the top trajectory. The local comparison alone is not an estimate of the entire activity on the smaller torus; that bound follows from its own continuation estimate. The vacuum is a common scalar and satisfies the top bound at this same level. Thus (64) follows. At a smaller-volume stop the couplings need not vanish; the zeros defining $P_N(e)$ are imposed at k_{top}.

The fixed-point map on the top torus is determined by $(N, e, \bar{m})$, the fixed construction parameters and photon coordinate basis, and the top primitive families constructed from them. The selected top side is itself a function of $e, \bar{m}$ and these parameters. The smaller side R enters no part of this definition. Nor does a twist: the twist is a harmonic part of a connection inserted in the density identity, whereas the coupling maps are maps of entire local activity families and contain no such insertion. The full forward activity recursion retains the external slot as a variable, whereas the fixed-point map uses its restriction to zero. No external connection or probe is inserted when selecting the pair. This proves the asserted independence within the fixed construction.

For a cutoff on which the pair is defined, (65) is its initial mass bound on the top torus. At the other finitely many cutoffs it holds because the assigned mass is zero. Finally $h \geq 1$ implies $e \leq \bar{e}_1 \leq 1/3$, so $q \leq 1/3$ and its positive power is below 1. The sequence is real in both cases. The sign of $\boldsymbol{\mu}_N$ is not fixed by this estimate; the lattice construction admits a real mass. □

On the reference torus, for $N \geq N_{\mathrm{ref}}(e)$, the outputs needed at the terminal scale are

$$a_* \in \{h/L, h/L^2\}, \qquad |m_*| \leq \tfrac{1}{8} e_*^{3/4-8\varepsilon}, \qquad \|E_*\|_* \leq \|\mathscr{E}_*\|_*^x \leq 4 e_*^{1/4-6\varepsilon}. \tag{66}$$

The primitive estimates along this flow have the same fixed constants on every torus and at every cutoff. The remaining terminal density and partition estimates concern the integration of these fields.

Proof of Theorem A. Proposition 6.1 gives the stopping levels and thresholds; (54) gives the reference terminal scales. On the top torus, theorem C selects the unique controlled pair with both terminal couplings zero. Theorem D transfers this pair and its scalar bounds to each eligible smaller torus. Its proof gives the stated local agreement of both activity families, while proposition 7.1 supplies their full norm bounds. The initial mass estimate and its zero extension are (65). The same common-pair proof establishes independence from the smaller side, twist and external insertion. □

Appendix A. The local formulas

This appendix recalls the finite expressions used in the comparison. Their construction and estimates are the input results of section 3. In the formulas below, k is fixed and $B = A + \mathscr{C}$.

A.1. **Extraction and reconstruction.** The extraction maps of [NC, §3.1] use the ordinary quadratic part

$$\mathscr{E}^{[2,0]}(X; A, \mathscr{C}) = \int \mathrm{d}x\, \mathrm{d}x'\, \bar{\psi}(x)^T K_X(A, \mathscr{C}; x, x') \psi(x').$$

For a small X, choose one lift $\widetilde{X}$ and define

$$a_0(\mathscr{E}, X) = \mathrm{Vol}(X)^{-1} \mathscr{E}_{00}(X; 0, 0), \qquad a_2(\mathscr{E}, X) = \mathrm{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}x \int_{\widetilde{X}} \mathrm{d}x'\, K_X(0, 0; x, x'). \tag{67}$$

Both cluster labels are averaged with measure $\eta_k^3/48$. The second expression is independent of the lift: simultaneous translation by a period multiplies its two arguments by the same antiperiodic sign, whose product is one.

For an M-cube $\square$ put

$$\varepsilon_{\mathrm{v}}(\mathscr{E}, \square) = - \sum_{\substack{X \supset \square \\ X \text{ small}}} a_0(\mathscr{E}, X), \qquad M(\mathscr{E}, \square) = - \sum_{\substack{X \supset \square \\ X \text{ small}}} a_2(\mathscr{E}, X). \tag{68}$$

Translations by $M\mathbb{Z}^3$ identify the small-polymer sets over different cubes. They leave the two extracted quantities unchanged. Thus the two sums do not depend on $\square$. Write their values as $\varepsilon_{\mathrm{v}}(\mathscr{E})$ and $M(\mathscr{E})$.

The imported scalarity assertion is $M(\mathscr{E}) = m(\mathscr{E})1$. On a small polymer the normalized remainder is [NC, (17)]

$$(R\mathscr{E})(X; A, \mathscr{C}) = \mathscr{E}(X; A, \mathscr{C}) - a_0(\mathscr{E}, X)\, \mathrm{Vol}(X) - \int_X \mathrm{d}x\, \bar{\psi}(x)^T a_2(\mathscr{E}, X) \psi(x). \tag{69}$$

On other polymers it leaves the activity unchanged. The operation $\Theta_{X,B}$ is the shortest-chain quadratic adjustment of [NC, §3.2]. It averages over the shortest chains of M-cubes in X, using the same six-path transports and pair coefficients d_X^α. We use the resulting $(R\mathscr{E})^{\mathrm{nat}}$ from that definition.

Let Δ_y be the unit block centered at y. For $c \in \Delta_y$, let $c_j(c)$ be the center of its block of side $L^j\eta_k$. The averaging phase and its normalized average are

$$\varphi_A(y,c) = \exp\left(ie_k\eta_k \sum_{j=0}^{k-1} (\tau A)(c_{j+1}(c), c_j(c)) \right), \qquad (\mathcal{Q}_k(A)f)(y) = \frac{\eta_k^3}{48} \sum_{c\in\Delta_y, g} \varphi_A(y,c) f(c,g).$$

The reverse is the weighted transpose $\mathcal{Q}_k(-A)^T$. For $y \in Y$ the two reconstruction rows of [NC, Theorem B] are as follows, where $c_3 = \pi x^{(3)}$.

$$\begin{aligned}(T_Y^{(1)}(A)\psi)(y) &= \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{(1+\mu/b_k)\psi(x) \\ &\qquad - (b_k\eta_k)^{-1} \sum_{i:\, x^{(i)} \text{ over } Y} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\}, \\ (T_Y^{(2)}(A)\chi)(y) &= -b_k^{-1}\eta_k^{\alpha-1} \int_{\Delta_y} \mathrm{d}x\, 1[c_3 \notin Y]\varphi_A(y,c) M_3(x)\chi(x,x^{(3)}).\end{aligned} \tag{70}$$

Rows outside Y are zero. The first row contains ordinary legs only over Y. The second row contains precisely the type-3 pairs leaving Y, which are admissible boundary pairs of length η_k. Their coefficients read A only on bonds with an endpoint in Δ_y. The barred rows are $\bar{T}_Y^{(i)}(B) = \mathsf{C}^{-1}T_Y^{(i)}(-B)\mathsf{C}$. The substitution $\mathrm{sub}_{Y,B}$ replaces independent unit generators $\Psi, \bar{\Psi}$ by these ordinary and pair rows and fixes the old generators. This specifies the homomorphism used below.

A.2. **The determinant primitive and the first vertex.** Use the unprimed polymer resolvents $S_{k,y}(X,B)$ of [FB, §4.3]. Write $\mathcal{B}_{k,y} = B_{k,y}$ for the finite block resolvent compared in section 5.3, and $b_L = L^{-1}$. At zero $F_0 = D_B + \mu + b_L J P_L J^*$ and $\Gamma_0 = F_0^{-1}$. The determinant primitive is [FB, (76)]:

$$\widetilde{E}_k(X,A) = -ib_k^2 \int_0^\infty \mathrm{d}y\ \mathrm{Tr}[\gamma_5 \mathcal{B}_{k,y}\mathcal{Q}_k S_{k,y}(X,A)\mathcal{Q}_k^\sharp \mathcal{B}_{k,y}] \quad (k \geq 1), \tag{71}$$

$$\begin{aligned}\widetilde{E}_0(X,A) &= \int_1^\infty \frac{\mathrm{d}y}{y}\, \mathrm{Tr}[F_0 S_{0,y}(X,A)] - i\int_0^1 \mathrm{d}y\ \mathrm{Tr}[\gamma_5 S_{0,y}(X,A)], \\ E_k^d(X,A) &= \tfrac{1}{4}(\widetilde{E}_k(X,A) + \widetilde{E}_k(X,-A) + \overline{\widetilde{E}_k(X,\overline{A})} + \overline{\widetilde{E}_k(X,-\overline{A})}), \\ E_k^{\det}(X,A) &= E_k^d(X,A) - E_k^d(X,0).\end{aligned}$$

The traces use unit counting measure for coarse spinors and the chamber measure $1/48$ at level zero. Absolute convergence and the reblocked gain are part of the determinant input.

At zero write $q = e_0$ for the running charge. The local vertex of [NL, (47)] is

$$\begin{aligned}\mathcal{V}_0(Z,\square) = \frac{1}{2}\int_\square \mathrm{d}x \Big\{ &\bar{\psi}(x)^T M_3(x)(e^{iqZ(c,c_3)} - 1)[\psi(x) + \chi(x,x^{(3)})] \\ &+ [\bar{\psi}(x) + \bar{\chi}(x,x^{(3)})]^T M_3(x^{(3)})(e^{-iqZ(c,c_3)} - 1)\psi(x)\Big\},\end{aligned} \tag{72}$$

where $x = (c,g)$, $c_3 = c + ge_1$ and $\int_\square \mathrm{d}x = 48^{-1}\sum_{c\in\square}\sum_g$. The pair $(x, x^{(3)})$ has $d_X = 1$ and is admissible for $\square$ of the second kind. Thus the vertex has ordinary legs over the cube and precisely the boundary pair legs allowed in the activity algebra. Its coefficients are entire functions of Z on the bonds meeting the cube.

A.3. **The two finite localizations.** The definitions in this subsection are those of [NL, §5]. For $\zeta \in \mathrm{Box}_k$ put

$$Z(s') = H_k^{\mathrm{ph}}(s') C C_k^{1/2}(s')\zeta. \tag{73}$$

The radii, with $e = e_k$, are

$$r_1 = \tfrac{1}{2}e^{-3/4+9\varepsilon/2}, \qquad r_2 = e^{-3/4+7\varepsilon}, \qquad r_4 = e^{-1/4+3\varepsilon}. \tag{74}$$

The imported contour bounds give $2 \le r_2, r_4 \le r_1$ and $A + tZ \in \widetilde{R}_k$, $B + tZ \in \widetilde{G}_k$ for $|t| \le r_1$. The arguments of $H_k^{\#}$ are independent pairs $(\Phi, \bar{\Phi})$; in $H_k^{\#}\Phi$ the single symbol denotes this pair. Write $\psi^{\#} = (\psi, \chi; \bar{\psi}, \bar{\chi})$ for the ordinary, pair, barred ordinary and barred pair generators, respectively. Then

$$\begin{aligned} H_k^{\#}(s'', B)\Phi = (&H_k(s'', B)\Phi, \delta_{\alpha,B} H_k(s'', B)\Phi; \\ &\bar{H}_k(s'', B)\bar{\Phi}, \delta_{\alpha,-B}\bar{H}_k(s'', B)\bar{\Phi}), \\ \bar{H}_k(s'', B) &= \mathsf{C}^{-1} H_k(s'', -B)\mathsf{C}. \end{aligned}$$

Define $J_k^{\#}(s'', B, Z) = H_k^{\#}(s'', B + Z) - H_k^{\#}(s'', B)$ and

$$\widetilde{\Gamma}_k(s'', B)(W, \bar{W}) = (\Gamma_k(s'', B)W, \bar{W}), \qquad W_k^{\#}(s'', B) = H_k^{\#}(s'', B)(\Gamma_k(s'', B)W, \bar{W}).$$

The notation $J_k, \bar{J}_k$ denotes the components without Hölder quotient; $\mathrm{Wc}, \overline{\mathrm{Wc}}$ denotes the corresponding components of $W_k^{\#}$. The auxiliary generators W have weight $\mathsf{w}_W = 1$ for $k \ge 1$ and $\sqrt{48}$ for $k = 0$.

In the following localization, cubes and polymers are sets of photon sites. Containment refers to all their lattice sites, and two cubes meet when they share a lattice site. Let $P_0(X)$ be the parameter cubes contained in X. For a set P of parameter cubes disjoint from $P_0(X)$, let $Y'(X, P)$ be the union of the M-cubes meeting X or a cube of $P_0(X) \cup P$. For a continuous analytic family $G(s, X)$ define

$$\mathcal{T}_P G(X) = \int_{[0,1]^P} \mathrm{d}s_P \, \partial_P G(s, X)\big|_{s=1 \text{ on } P_0(X),\, s=0 \text{ outside } P_0(X)\cup P}, \tag{75}$$

$$\mathrm{Loc}\, G(Y) = \sum_{(X,P):Y'(X,P)=Y} \mathcal{T}_P G(X). \tag{76}$$

For a function analytic near the closed disk $|t| \le r$, $r > 1$, write

$$\mathcal{C}_r f = \frac{1}{2\pi i}\int_{|t|=r} \frac{f(t)}{t(t-1)}\,\mathrm{d}t.$$

This is $f(1) - f(0)$ and satisfies $\|\mathcal{C}_r f\| \le (r-1)^{-1}\sup_{|t|=r}\|f(t)\|$. Let $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(4)$ and set $m' = m + m(\mathscr{E})$, $\mathscr{E}^r = R\mathscr{E}$. For $k \ge 1$ define

$$G_1(s, X) = \mathcal{C}_{r_1}\langle\bar{\Psi}, M_k(X, B + tZ(s'))\Psi\rangle_1, \tag{77}$$

$$G_2(s, X) = \mathcal{C}_{r_2}\,\mathscr{E}^r(X; A + tZ(s'), \mathscr{C}, \psi^{\#} + tJ_k^{\#}(s'', B, Z(s'))\Psi), \tag{78}$$

$$G_3(s, \square) = \mathcal{C}_{r_2}(-m')\langle\bar{\psi} + t\bar{J}_k(s'', B, Z(s'))\bar{\Psi}, \psi + tJ_k(s'', B, Z(s'))\Psi\rangle_{\square}. \tag{79}$$

The symbols $M_k(X, \cdot)$ denote the local Dirac primitives. A function defined for one cube is zero on the other polymers. Put

$$E_i^*(Y) = \mathrm{Loc}\, G_i(Y), \qquad E_i(Y) = \mathrm{sub}_{Y,B}\, E_i^*(Y), \qquad E_{\le 3} = \sum_{i=1}^{3} E_i.$$

The substitution was defined above. At $k = 0$ there is no Ψ and no substitution: $G_1(s, \square) = \mathcal{V}_0(Z(s'), \square)$, $G_2 = G_3 = 0$, and $E_{\le 3} = \mathrm{Loc}\, G_1$. All parameter kinds occur in this first localization.

For the second localization, which uses only π_k, let $v = W_k^{\#}(s'', B)$ and define

$$G_{\le 3}^W(s'', X) = E_{\le 3}(X; A, \mathscr{C}, \zeta, \psi^{\#} + v), \tag{80}$$

$$G_4(s'', X) = \mathcal{C}_{r_4}\mathscr{E}^r(X; A, \mathscr{C}, \psi^{\#} + tv), \tag{81}$$

$$G_5(s'',\square) = \mathcal{C}_{r_4}(-m')\langle \bar\psi + t\overline{\mathrm{Wc}}, \psi + t\mathrm{Wc}\rangle_\square, \tag{82}$$

$$F(Y) = \mathrm{Loc}(G^W_{\le 3} + G_4 + G_5)(Y). \tag{83}$$

At $k=0$ the term G_4 is zero and $m'=m$.

A.4. **Integration, logarithm and reblocking.** For the even activity F just defined, set

$$K(Y) = \sum_{\substack{\{X_i\}\ \text{distinct} \\ \bigcup_i X_i = Y}} \prod_i (\exp F(X_i) - 1), \qquad K(\varnothing) = 1,$$

$$H(Y) = \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}_k(\zeta)\, K(Y),$$

$$E^{\#}_k(Y) = \sum_{n\ge 1} \frac{1}{n!} \sum_{\bigcup_i Y_i = Y} u^T(Y_1,\dots,Y_n) \prod_i H(Y_i).$$

Here $\widehat{\mu}_k$ is the product of the normal laws restricted and normalized on $[-p_{0,k}, p_{0,k}]$. The measure $\mu_I(W)$ is the normalized identity-covariance Grassmann measure: the covariance kernel is the inverse mode weight. The coefficient u^T is the sum of $(-1)^{|E(G)|}$ over connected spanning subgraphs of the incompatibility graph of the tuple. Two polymers are incompatible when their union is connected; in particular repetitions are incompatible. Absolute convergence in the weighted coefficient norm is supplied by [NL, Proposition 6.1].

For an outgoing polymer Y, put

$$E^{*}_k(Y; A', \mathscr{C}') = \sum_{\bar X^L = LY} S_L E^{\#}_k(X; A'_L, \mathscr{C}'_L), \qquad A'_L(b) = L^{-1/2} A'(b/L).$$

The bar denotes the union of LM-cubes containing the M-cubes of X. The map S_L multiplies ordinary and pair legs by L^{-1} and $L^{-1-\alpha}$, respectively. The same sum defines $\mathcal{L}$ on any input activity. The scaled-domain inclusion used in comparison is

$$(A', \mathscr{C}') \in \widetilde{R}_{k+1} \times \mathfrak{X}_{k+1} \quad \Longrightarrow \quad (A'_L, \mathscr{C}'_L) \in \widetilde{R}^-_k \times \mathfrak{X}_k; \tag{84}$$

it is [NC, (29)] under the assumed numerical conditions.

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